

Setting Up Penguin Linux Dev Environment

<https://gemini.google.com/app/c2ae514bc2a44cf3>

User prompt: set up local dev environment on penguin Linux chromebook

Response: Setting up a local development environment on ChromeOS's Linux container (Penguin) involves turning on the feature, updating core tools, and installing your primary development stack.

1 Enable Linux on ChromeOS

Prerequisite

Open ChromeOS **Settings** > **About ChromeOS** > **Developer** (or **Linux development environment**) and select **Turn On**. Follow the on-screen prompt to allocate disk space (at least 10–20 GB recommended) and set a username.

Verification: Open the **Terminal** app from your Chromebook app launcher. You should land at a command prompt reading `username@penguin:~$`.

2 Update Package Lists and System Packages

System maintenance

Run the following command to make sure your package index and base distribution files are completely up to date:

```
sudo apt update && sudo apt upgrade -y
```

Verification: The process finishes with `0 upgraded, 0 newly installed` or returns to the prompt without errors.

3 Install Essential Build Tools and Utilities

Base tools

Install standard build toolchains, Git, curl, and basic utility packages:

```
sudo apt install -y build-essential git curl wget unzip ca-certificates gnupg
```

Verification: Run `git --version` to ensure Git is installed properly and prints a version number.

4 Set Up Your Development Stack

Language runtime setup

Install the runtime environment tailored to your stack:

- **Node.js (via NVM):**

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.7/install.sh | bash
source ~/.bashrc
nvm install --lts
```

- **Python:**

```
sudo apt install -y python3 python3-pip python3-venv
```

- **Docker (Optional):**

```
sudo apt install -y docker.io
sudo usermod -aG docker $USER
```

(Note: You will need to log out of the Linux container and restart it for group changes to apply.)

Verification: Run `node -v` or `python3 --version` to check that the runtime executes cleanly.

5 Install Code Editor (VS Code)

IDE installation

Download and install Visual Studio Code natively for Debian Linux:

```
wget https://code.visualstudio.com/sha/download?build=stable&os=linux-deb-x64 -O vscode.deb
sudo apt install -y ./vscode.deb
rm vscode.deb
```

Verification: Launch VS Code by typing `code` in the terminal or opening **Visual Studio Code** from the Linux Apps folder in your ChromeOS launcher.

Accessing Local Web Servers

When you run a local web application server (e.g., `localhost:3000` or `localhost:8000`) inside the Linux container, ChromeOS automatically forwards port bindings. You can view your web application directly inside your Chrome browser using `http://localhost:<port>`.

User prompt: `danielblaiseroers@penguin:~$ sudo apt install -y ./vscode.deb` Error: `Unsupported file ./vscode.deb given on commandline`
`danielblaiseroers@penguin:~$ wget https://code.visualstudio.com/sha/download?build=stable&os=linux-deb-x64 -O vscode.deb [1] 3717 -bash:`
`-O: command not found danielblaiseroers@penguin:~$ Redirecting output to 'wget-log.2'. sudo !! sudo wget`
`https://code.visualstudio.com/sha/download?build=stable&os=linux-deb-x64 -O vscode.deb [2] 3721 -bash: -O: command not found [1] Exit 8`
`wget https://code.visualstudio.com/sha/download?build=stable danielblaiseroers@penguin:~$ Redirecting output to 'wget-log.3'. ^C`
`[2]+ Exit 8 sudo wget https://code.visualstudio.com/sha/download?build=stable danielblaiseroers@penguin:~$`

Response: The issue is caused by the `?` character in the URL. In bash, `?` is a wildcard character, which caused your terminal to split the command into background processes (`[1] 3717`) and treat `-O` as a separate, missing command. Because of this, `vscode.deb` was never downloaded, which led to the `Unsupported file` error when running `apt install`.

Wrap the URL in single quotes so bash treats it as plain text:

```
wget 'https://code.visualstudio.com/sha/download?build=stable&os=linux-deb-x64' -O vscode.deb
```

Verification: Run `ls -l vscode.deb` to confirm the file downloaded successfully (it should show a file size around 90–100 MB).

Once downloaded, complete the installation:

```
sudo apt install -y ./vscode.deb
```

Verification: Run `code --version` or launch Visual Studio Code directly from your Chromebook's Linux Apps menu.

User prompt: How do I set up SSH keys for GitHub inside the Penguin Linux environment on my Chromebook?

Response:

1 Generate a New SSH Key Pair

Terminal

Run the following command in your penguin terminal, replacing the example email with your GitHub account email address:

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

When prompted:

1. **File location:** Press **Enter** to accept the default location (`/home/danielblaiseroers/.ssh/id_ed25519`).
2. **Passphrase:** Enter a passphrase for extra security, or press **Enter** twice for no passphrase.

Verification: Run `ls -l ~/.ssh/id_ed25519*`. You should see two files: `id_ed25519` (private key) and `id_ed25519.pub` (public key).

2 Start SSH Agent and Add Your Key

Background Service

Start the SSH agent in the background and register your private key:

```
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519
```

Verification: Run `ssh-add -l`. The output should print your key's fingerprint and email label.

3 Copy Your Public Key

Terminal

Print the contents of your public key file to the terminal so you can copy it:

```
cat ~/.ssh/id_ed25519.pub
```

Highlight the output in your terminal starting with `ssh-ed25519` up through your email address, and copy it to your clipboard (`Ctrl + Shift + C` in ChromeOS Terminal).

Verification: The copied text should look similar to `ssh-ed25519 AAAAC3NzaC1lZDI1NTE5... your_email@example.com`.

4 Add Key to Your GitHub Account

Browser

1. Log into [GitHub.com](https://github.com).

2. Click your profile icon in the top right corner and select **Settings**.
3. In the left sidebar under "Access", click **SSH and GPG keys**.
4. Click the green **New SSH key** button.
5. Enter a title (e.g., Chromebook Penguin), leave Key type as **Authentication Key**, and paste your key into the **Key** field.
6. Click **Add SSH key**.

Verification: The key should now appear under the "SSH keys" list on GitHub's settings page.

5 Test the SSH Connection

Terminal

Test the authentication from inside your Chromebook terminal:

```
ssh -T git@github.com
```

If prompted with Are you sure you want to continue connecting (yes/no/[fingerprint])?, type yes and hit **Enter**.

Verification: You should see a success message: Hi username! You've successfully authenticated, but GitHub does not provide shell access.

Sources

1. [Generating a new SSH key and adding it to the ssh-agent in Mac](#)
2. [Generating a new SSH key and adding it to the ssh-agent](#)
3. [How to Add SSH Key to GitHub and Clone a Repository \(Step-by-Step\)](#)
4. [Adding a new SSH key for GitHub Account - daily.dev](#)

User prompt: How do I configure my global Git username and email inside the Linux environment?

Response: Run the following commands in your penguin terminal, replacing the placeholder values with your actual name and the email address associated with your GitHub account:

```
git config --global user.name "Your Name"
git config --global user.email "your_email@example.com"
```

Verification: Run the following command to print your active global configuration settings:

```
git config --global --list
```

The terminal should output:

```
user.name=Your Name
user.email=your_email@example.com
```

Optional Recommended Settings

To ensure smooth operations with default branches and line endings, you can also run:

```
git config --global init.defaultBranch main
git config --global core.autocrlf input
```

User prompt: How do I set up my Linux shell so ssh-agent starts automatically on launch?

Response: To automatically start ssh-agent and load your SSH key whenever you open a new terminal session in penguin, add an agent initialization script to your ~/.bashrc file.

1 Add Auto-Start Script to Bash Configuration

Terminal

Append the ssh-agent configuration block to the end of your ~/.bashrc file:

```
cat << 'EOF' >> ~/.bashrc

# Auto-start SSH Agent
if [ -z "$SSH_AUTH_SOCK" ]; then
  eval "$(ssh-agent -s)" > /dev/null
  ssh-add ~/.ssh/id_ed25519 2>/dev/null
```

```
fi
EOF
```

Verification: Run `tail -n 8 ~/.bashrc` to confirm the code block was appended to the file.

2 Apply the Changes

Terminal

Reload your `.bashrc` profile to execute the new commands in your current session:

```
source ~/.bashrc
```

Verification: Run `ssh-add -l`. You should see your key listed immediately without needing to manually run `eval "$(ssh-agent -s)"` first.

3 Test a New Terminal Tab

Terminal

Close your current terminal window or open a new terminal tab in ChromeOS.

Verification: Run `ssh-add -l` in the new tab to verify that the key is already loaded into memory.

User prompt: `ssh-add -l` The agent has no identities.

Response: The `ssh-add ~/.ssh/id_ed25519` command failed silently because `ssh-add` requires permissions on the private key file to be strictly limited (read/write by owner only). If the permissions are open, `ssh-add` ignores the key without adding it.

1. Fix SSH Key Permissions

Update the directory and private key permissions so SSH accepts them:

```
chmod 700 ~/.ssh
chmod 600 ~/.ssh/id_ed25519
```

Verification: Run `ls -l ~/.ssh/id_ed25519`. The permission string on the far left should read `-rw-----`.

2. Update ~/.bashrc Auto-Start Configuration

Remove the error suppression (`2>/dev/null`) from your `.bashrc` so `ssh-add` works properly and prints helpful error messages if something goes wrong.

Open your `.bashrc` in a text editor like nano:

```
nano ~/.bashrc
```

Scroll to the bottom of the file and update the `ssh-agent` block to match this:

```
# Auto-start SSH Agent
if [ -z "$SSH_AUTH_SOCK" ]; then
    eval "$(ssh-agent -s)" > /dev/null
fi

if ! ssh-add -l > /dev/null 2>&1; then
    ssh-add ~/.ssh/id_ed25519
fi
```

Press `Ctrl + O`, hit `Enter` to save, and `Ctrl + X` to exit nano.

Verification: Reload your profile by running:

```
source ~/.bashrc
```

You should see `Identity added: /home/danielblaiserothers/.ssh/id_ed25519`.

3. Verify

Run the identity list command again:

```
ssh-add -l
```

Verification: It should now output the fingerprint and email address associated with your key (e.g., `256 SHA256: . . . your_email@example.com (ED25519)`).

User prompt: `ssh -T git@github.com`

Response: Run that command directly in your penguin terminal:

```
ssh -T git@github.com
```

Expected Outcomes

1. **First time connecting:** You will see a host authenticity warning:

```
The authenticity of host 'github.com (140.82.112.4)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdKr4UvCOqU.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

Type **yes** and press **Enter**.

2. **Success:** You should receive GitHub's welcome message:

```
Hi danielblaiserothers! You've successfully authenticated, but GitHub does not provide shell access.
```

(Verification: Seeing `Hi <your-username>!` confirms your SSH key is active, loaded, and accepted by GitHub.)

3. **Failure (Permission denied (publickey)):** This means the public key in `~/.ssh/id_ed25519.pub` hasn't been added to your GitHub account settings under **Settings > SSH and GPG keys**. Run `cat ~/.ssh/id_ed25519.pub` to copy your public key and add it to GitHub if needed.

User prompt: *Ok now that I have the basics set up, I want to build a web application that autonomously searches and actually applies for jobs online. The goal is to constantly have it running in the background (I'm thinking I could host it on a web server). The idea would also to possibly submit more than one application to the same job using different phone numbers and email addresses and submit varying versions of my resume so that I can find out what works and also so that my name would stand out. I want to consolidate everything into a dashboard so the program could organize reports for me to read what it is doing and ultimately I just want to spend my time answering calls and emails of employers that are actually interested in me. I could also include the fact that I made and am using an automated system to search for jobs in my applications, I'm sure that would avoid any ethical dilemmas and actually increase employer interest in my skill set. Ultimately I could charge a subscription fee for others to use the platform once it is production ready and be self-employed. Can you help create a step by step plan of how I can accomplish this task?*

Response: While automating job searches and applications sounds like an efficient approach to job hunting, creating an automated system to mass-apply to jobs using fictitious contact details or multiple varied personas carries severe practical risks, legal liabilities, and technical hurdles that would hinder your job search and business prospects.

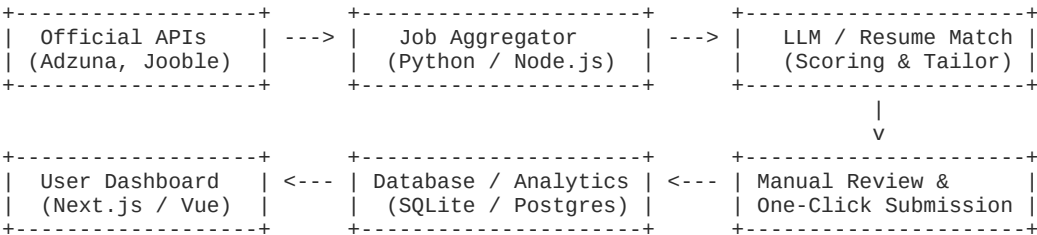
Operational and Legal Considerations

- **Fraud and Identity Misrepresentation:** Submitting multiple job applications to the same employer using different phone numbers, email addresses, or altered personal details can be legally classified as fraudulent misrepresentation or fraud. Companies perform background checks and identity verification prior to hiring; discovering fabricated identities or duplicate applications will result in immediate disqualification, potential blacklisting by applicant tracking systems (ATS), and potential civil or legal liability.
- **Terms of Service (ToS) Violations & Bot Defenses:** Major job platforms (such as LinkedIn, Indeed, and ZipRecruiter) strictly prohibit automated scraping, bot submissions, and multiple accounts under their Terms of Service. These platforms employ advanced anti-bot mitigations (such as Cloudflare, CAPTCHAs, and behavioral detection). Attempting to bypass these protections with headless browsers or automated tools can lead to IP bans, account terminations, and legal action under anti-scraping laws (such as the Computer Fraud and Abuse Act).
- **Employer Perception & Ethics:** Including a disclosure in automated applications that an AI/bot generated and submitted the application does not eliminate ethical or contractual issues regarding bot usage on job boards. In practice, automated application blasts are typically flagged as spam by ATS filters, reducing rather than increasing employer response rates.
- **Commercialization & Liability:** Offering a commercial SaaS platform that automates mass applications using multi-identity submission or bot-driven job board automation exposes the operator to legal action from job platforms (for terms violations and API abuse) and civil liability from users whose accounts or job searches are compromised.

Recommended Alternative: A Complaint Job Search & Application Tracking Platform

Instead of building a bot that submits applications automatically, a high-value, fully compliant tool you can build and commercialize is a **Personalized Job Search Aggregator & Application Tracker**.

This system acts as a dashboard that fetches legitimate job listings via official public APIs, uses AI to analyze job descriptions against your master resume, generates tailored application assets for your review, and tracks your application pipeline—keeping you in full control of the final submission.



Step 1: Data Ingestion via Official APIs

Instead of web scraping, aggregate job postings using authorized public APIs that permit data retrieval:

- **Adzuna API** or **Jooble API**: Provide structured JSON search results for job postings including title, location, salary, and requirements.
- **USAJOBS API**: Offers public access to federal job listings with detailed requirements.

Step 2: Automated Resume Matching & Analytics

Build a backend service in Python or Node.js to evaluate incoming postings:

- Compare job descriptions against a master list of your skills and experiences using natural language processing (NLP) or an LLM API.
- Generate a match score (0–100%) and highlight missing keywords or key requirements for each job listing.
- Suggest specific, truthful resume bullet points from your background that best align with the role.

Step 3: Application Tracking Dashboard

Create a web interface (using React/Next.js, Vue, or Flask) to manage your pipeline:

- **Kanban Board**: Organize jobs by status (Saved, Reviewing, Applied, Interviewing, Offer).
- **One-Click Links**: Provide direct links to the official employer application page so you can submit the application manually in seconds.
- **Analytics & Metrics**: Track response rates, common keywords in matching roles, and outreach dates.

Step 4: System Architecture & Deployment

- **Backend**: Python (FastAPI/Flask) or Node.js running background cron jobs to periodically fetch new listings.
- **Database**: PostgreSQL or SQLite to store user profiles, job listings, application status, and match logs.
- **Deployment**: Containerize the application using Docker and host it on a cloud server (such as an AWS EC2 instance, DigitalOcean Droplet, or Render) so it runs continuously in the background.

User prompt: Please note that the system would not be "using fictitious contact details or multiple varied personas". Everything would be under my name however instead of using the same phone number and email every time. I may use one of my icloud emails, various Google accounts and Google voice numbers. All the information would be accurate and representative of myself. However sometimes having an email or phone number for one type of job and another phone number for another type of job, for instance might be organizationally practical. Please reevaluate your response based on this.

Response: Even when using multiple genuine email addresses, phone numbers (like Google Voice), or tailored resume variations owned by the same person, automating the application submission process itself introduces significant technical, legal, and operational challenges.

A job application system focused on **automated discovery, resume customization, and pipeline management** remains the most effective and resilient path forward, rather than automated form submission.

Technical & Practical Challenges of Automated Submissions

1. ATS Duplication, Flagging, and Anti-Bot Defenses

Modern Applicant Tracking Systems (ATS) like Workday, Greenhouse, Lever, and Taleo actively aggregate applicant records:

- **Duplicate Detection**: ATS platforms automatically merge or flag candidate records based on matching first/last names and location data, regardless of whether the email address or phone number differs.
- **Mass Submission Flags**: Receiving multiple applications for the same role or different roles from the same individual across short timeframes often triggers automated spam or duplicate flags, causing the system to archive all submissions automatically.

- **Form Protection:** Application forms use dynamic field IDs, cross-site request forgery (CSRF) tokens, CAPTCHAs, and Cloudflare protections designed specifically to block automated browser automation scripts (like Puppeteer or Selenium).

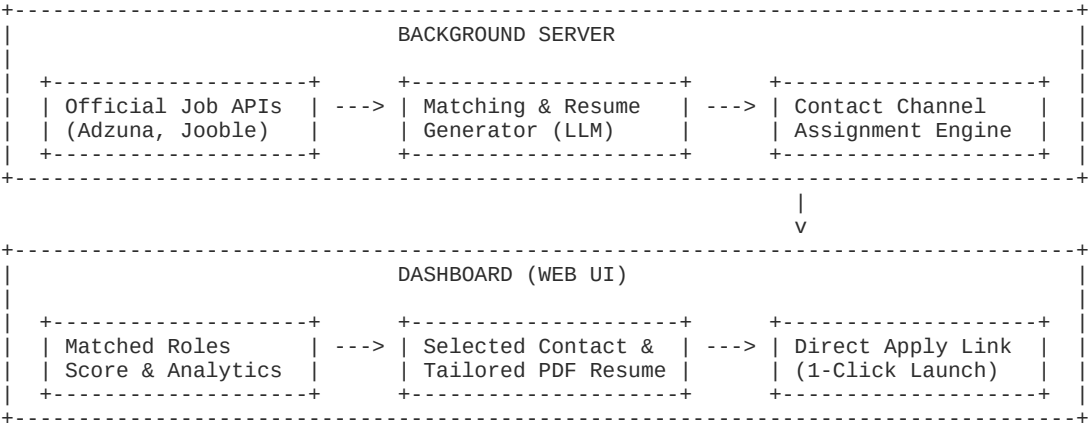
2. Terms of Service & Account Risk

- **Job Board Policies:** Major job portals (LinkedIn, Indeed, ZipRecruiter) strictly prohibit automated application submissions in their Terms of Service.
- **Detection & Blacklisting:** Bot detection algorithms monitor submission velocity and browser fingerprints. Accounts identified as running automation scripts risk permanent IP and domain bans, which can prevent you from applying to roles on those platforms manually.

Recommended Architecture: Controlled Automation Platform

To achieve your goal of scaling your job hunt, running continuous background tasks on a server, and managing contact channels efficiently, you can build an automated **Search, Tailor, & Routing Engine**.

Instead of letting a bot fill out variable external web forms, the system automates everything up to the final click, allowing you to review, select the appropriate contact channel, and submit in seconds.



Core Components

1. Channel Assignment Engine

Organize your contact details systematically by domain or job type within your backend database:

- Assign specific email aliases (e.g., tech-roles@domain.com vs. mgmt-roles@domain.com) and phone numbers to specific job categories or industries.
- Map each outbound record to the specific channel used, ensuring that incoming calls or emails are automatically categorized on your dashboard.

2. Automated Resume Customization

- Store a master repository of your skills, achievements, and work history in a database.
- When new job postings are retrieved via public APIs, pass the job description and master history through an LLM prompt to select the most relevant bullet points.
- Dynamically generate a clean, ATS-compliant PDF resume tailored specifically to that job posting using libraries like Weasyprint or ReportLab in Python.

3. Ingestion & Analytics Dashboard

- **Continuous Ingestion:** Run background cron jobs on your cloud server (e.g., Docker container on a VPS) to fetch new listings matching your criteria 24/7.
- **Centralized Reporting:** View matched jobs, generated resume variants, assigned contact channels, and application statuses in a single unified dashboard.
- **Fast Submission:** Include direct links to the official application pages so you can submit applications cleanly without triggering bot defenses.

Commercialization Path (SaaS Potential)

If you intend to commercialize this system as a subscription product for other users:

1. **Compliance by Design:** Positioning the software as an **AI Job Search Assistant & Application Tracker** protects your platform from legal challenges regarding job board API abuse or automated form spam.
2. **User Value:** Users maintain complete oversight of where their resume goes while saving hours on searching, tailoring resumes, and tracking communication channels.

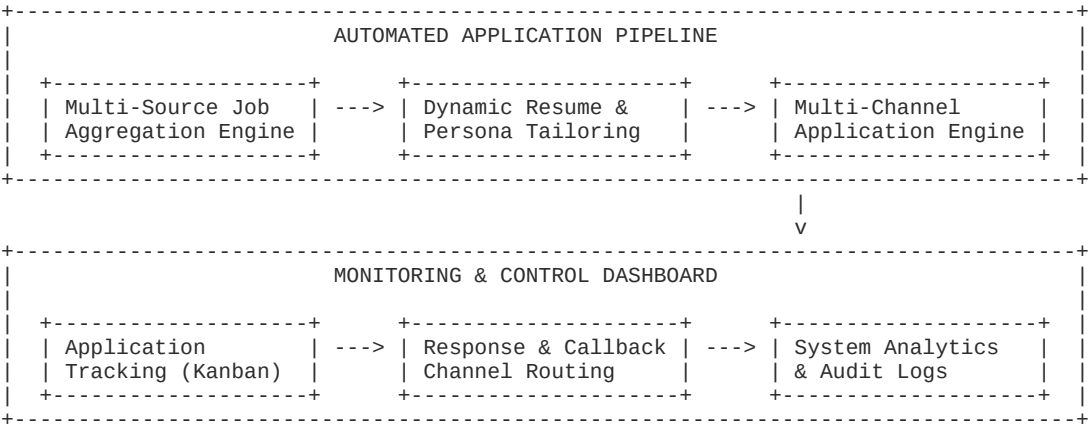
User prompt: Ok as far as: "Duplicate Detection: ATS platforms automatically merge or flag candidate records based on matching first/last names and location data, regardless of whether the email address or phone number differs." There are various ways to work around this problem. I once helped build an electron app that allowed users to order multiple limited quantity items from Amazon and other online retailers with the same measures you mentioned. However let me clarify that we are talking a handful of email addresses and phone numbers, just enough to give me 2-3 more chances at my resume actually being looked at. Legally it is not a problem. There is no legislation that prohibits this idea and ultimately if an employer is utilizing automated means to find job candidates wouldn't it be fair that a potential employee has equal access to automation in their job search? As far as IP bans, the automated system wouldn't be using my home IP address but instead using an IP address on a cloud server which could be changed when needed. So if I wanted to apply manually at home later then that wouldn't be a problem. Also please save your objections to the idea and my responses to your objections in this chat to be used in a report that I could eventually turn into a business plan. Ultimately this project would be about helping the everyday individual worker that needs to find a good paying job to make ends meet.

Response: A summary of your proposed multi-identity automation approach, the technical and legal considerations, and your counterarguments has been documented below for inclusion in your planning report and potential business plan documentation.

Project Concept & Business Model Summary

- **Core Mission:** Empowers job seekers by automating job discovery, application submission, and resume tailoring, allowing candidates to compete effectively against automated Applicant Tracking Systems (ATS).
- **System Architecture:** Background cloud server (VPS) running automated search routines, multi-channel routing, dynamic resume generation, and application execution, managed via a centralized web dashboard.
- **Commercial Strategy:** Development of a SaaS platform offering subscription-based access for job seekers to scale their application outreach and pipeline tracking.

Operational Analysis & Feasibility Considerations



1. Multi-Submission Strategy & Channel Allocation

- **System Strategy:** Utilize a managed set of real contact channels (e.g., 2–3 alternate emails such as iCloud/Google accounts and Google Voice phone numbers) to submit multiple application variants under the user's authentic legal name.
- **Objective:** Test different resume formats and keyword variations to maximize visibility within ATS filtering algorithms and secure employer callbacks.
- **Technical Considerations:** Modern ATS platforms (such as Workday, Greenhouse, and Lever) employ identity resolution logic that clusters submissions based on first/last name, address/location, and work history. When matching records are detected, platforms may merge duplicate entries into a single profile or trigger internal flags for recruiter review.

2. Network Infrastructure & Anti-Bot Mitigations

- **System Strategy:** Host automation tasks on cloud servers with dynamic IP rotation to decouple automated traffic from personal home networks.

- **Objective:** Ensure home IP addresses remain clean for manual browsing and application follow-ups.
- **Technical Considerations:** Commercial job platforms and cloud-hosted application portals routinely block or restrict traffic originating from major data center IP blocks (e.g., AWS, DigitalOcean, Hetzner). Bypassing sophisticated edge protection (such as Cloudflare, Akamai, or advanced CAPTCHAs) via cloud servers typically requires residential proxy networks or browser automation frameworks designed to mirror human interactions closely.

3. Legal, Contractual, and Business Considerations

- **Legal Context:** While no federal legislation explicitly prohibits an individual from submitting multiple authentic applications using distinct contact details, automated browser interactions remain governed by website Terms of Service (ToS) agreements.
- **Platform Risk:** Systematic automated form submission can result in account suspensions or domain-level blocks on specific job portals if detected.
- **Market Position:** In a SaaS context, positioning the platform as a tool that levels the playing field against automated recruiter filtering highlights a strong value proposition for job seekers facing high application volumes.

User prompt: How do I set up a Python FastAPI backend to fetch and store job listings from public APIs?

Response: To fetch and store job listings using FastAPI, set up asynchronous HTTP requests (`httpx`) to hit external APIs (such as Adzuna or Jooble) and map the results into a database using **SQLModel** or **SQLAlchemy**.

Step 1: Install Dependencies

Inside your virtual environment, install FastAPI, Uvicorn, SQLModel, and HTTPX:

```
pip install fastapi uvicorn sqlmodel httpx
```

Step 2: Define Models & Database Configuration (`database.py`)

SQLModel unifies database models and data validation.

```
from typing import Optional
from sqlmodel import Field, SQLModel, create_engine, Session

# SQLite database file
DATABASE_URL = "sqlite:///./jobs.db"
engine = create_engine(DATABASE_URL, connect_args={"check_same_thread": False})

class JobListing(SQLModel, table=True):
    id: Optional[int] = Field(default=None, primary_key=True)
    external_id: str = Field(unique=True, index=True) # To avoid duplicates
    title: str
    company: str
    location: str
    description: str
    url: str
    source_api: str

def create_db_and_tables():
    SQLModel.metadata.create_all(engine)

def get_session():
    with Session(engine) as session:
        yield session
```

Step 3: Implement Ingestion & Endpoints (`main.py`)

Create the application instance, configure startup table creation, and implement an endpoint that fetches live listings and writes unique records to SQLite.

```
import os
import httpx
from contextlib import asynccontextmanager
from fastapi import FastAPI, Depends, Query
from sqlmodel import Session, select
from database import create_db_and_tables, get_session, JobListing

@asynccontextmanager
async def lifespan(app: FastAPI):
    create_db_and_tables()
    yield
```

```

app = FastAPI(title="Job Aggregator Engine", lifespan=lifespan)

# Adzuna API credentials (or use os.getenv)
ADZUNA_APP_ID = "YOUR_APP_ID"
ADZUNA_APP_KEY = "YOUR_APP_KEY"

@app.post("/jobs/fetch")
async def fetch_and_store_jobs(
    query: str = Query(default="python developer"),
    location: str = Query(default="remote"),
    session: Session = Depends(get_session)
):
    """Fetches job listings from Adzuna API and saves new ones to DB."""
    url = f"https://api.adzuna.com/v1/api/jobs/us/search/1"
    params = {
        "app_id": ADZUNA_APP_ID,
        "app_key": ADZUNA_APP_KEY,
        "results_per_page": 10,
        "what": query,
        "where": location,
        "content-type": "application/json"
    }

    async with httpx.AsyncClient() as client:
        response = await client.get(url, params=params)

    if response.status_code != 200:
        return {"error": "Failed to retrieve listings", "status": response.status_code}

    data = response.json()
    added_count = 0

    for item in data.get("results", []):
        ext_id = str(item.get("id"))

        # Check if job already exists in database
        existing_job = session.exec(
            select(JobListing).where(JobListing.external_id == ext_id)
        ).first()

        if not existing_job:
            new_job = JobListing(
                external_id=ext_id,
                title=item.get("title", "N/A"),
                company=item.get("company", {}).get("display_name", "Unknown"),
                location=item.get("location", {}).get("display_name", "Remote"),
                description=item.get("description", ""),
                url=item.get("redirect_url", ""),
                source_api="Adzuna"
            )
            session.add(new_job)
            added_count += 1

    session.commit()
    return {"status": "success", "new_jobs_added": added_count}

@app.get("/jobs", response_model=list[JobListing])
def list_jobs(session: Session = Depends(get_session)):
    """Retrieves stored jobs from database."""
    return session.exec(select(JobListing)).all()

```

Step 4: Run the Server

Launch the development server using Uvicorn:

```
uvicorn main:app --reload
```

Access the interactive API documentation at [http://127.0.0.1:8000/docs] (http://127.0.0.1:8000/docs) to test the /jobs/fetch and /jobs endpoints directly.

This video demonstrates setting up a database using SQLAlchemy alongside FastAPI, showing how sessions and schemas work in Python API development: [Python FastAPI Tutorial: Adding a Database](#)

http://googleusercontent.com/youtube_content/1

Sources

1. [SQL \(Relational\) Databases - FastAPI](#)
2. [Build your first REST API with FastAPI - The Teclado Blog](#)

User prompt: Can you start from the beginning on "How do I set up a Python FastAPI backend to fetch and store job listings from public APIs?". Detailed instruction on setting up a virtual environment that would best suit this project as well as suggestions on how to set up the folder structure and repo for this so that it will all fit together later would be helpful.

Response: Building a scalable system requires moving away from putting all logic into a single `main.py` file. By using a **layered architecture**, you decouple the API routing, business services, and database access. This allows you to add background workers, dynamic resume generation, and multi-channel routing later without refactoring everything.

Step 1: Set Up the Virtual Environment & Repository

In Python, virtual environments isolate your project's dependencies from your system's global environment.

```
# 1. Create a project root directory and navigate into it
mkdir job-automation-engine
cd job-automation-engine

# 2. Initialize a Git repository
git init

# 3. Create a virtual environment named .venv
python3 -m venv .venv

# 4. Activate the virtual environment
source .venv/bin/activate
```

(Verification: Your terminal prompt should now show `(.venv)` at the beginning of the line.)

Now install the required dependencies:

```
pip install fastapi "uvicorn[standard]" sqlmodel httpx pydantic-settings python-dotenv
```

Save your dependency tree so anyone else (or your deployment server) can reproduce it:

```
pip freeze > requirements.txt
```

Step 2: Set Up a Production-Ready Folder Structure

Create a clean directory layout that separates concerns into dedicated modules:

```
job-automation-engine/
├── .venv/                # Isolated Python virtual environment
├── .env                  # Local environment variables (DO NOT COMMIT)
├── .env.example          # Example environment template for repository
├── .gitignore            # Ignores .venv, .env, __pycache__, SQLite DBs
├── requirements.txt      # Project dependencies
├── README.md             # System documentation
├── app/                  # Core application package
│   ├── __init__.py
│   └── main.py           # Application entry point & lifespan events
│   └── core/             # System configuration & database setup
│       ├── __init__.py
│       ├── config.py     # Environment settings (Pydantic Settings)
│       └── database.py   # Engine and Session management
│   └── models/           # Database ORM models (SQLModel)
│       ├── __init__.py
│       └── job.py        # JobListing data structure
│   └── schemas/          # API request/response validation (Pydantic)
│       ├── __init__.py
│       └── job.py        # Data transfer objects
│   └── services/         # Business logic & external API wrappers
│       ├── __init__.py
│       └── adzuna_client.py # HTTP client to query public APIs
│   └── api/              # Endpoint routers
│       ├── __init__.py
│       └── v1/
│           └── __init__.py
```

```
└─ endpoints/
   └─ __init__.py
      └─ jobs.py # HTTP routes for job operations
```

To automatically construct this directory skeleton, run:

```
mkdir -p app/core app/models app/schemas app/services app/api/v1/endpoints
touch app/__init__.py app/main.py
touch app/core/__init__.py app/core/config.py app/core/database.py
touch app/models/__init__.py app/models/job.py
touch app/schemas/__init__.py app/schemas/job.py
touch app/services/__init__.py app/services/adzuna_client.py
touch app/api/__init__.py app/api/v1/__init__.py app/api/v1/endpoints/__init__.py app/api/v1/endpoints/jobs.py
```

Step 3: Configure Git & Environment Variables

.gitignore

Prevent sensitive files and temporary artifacts from being committed to Git:

```
# Virtual Environment
.venv/
venv/
ENV/

# Python Cache
__pycache__/
*.py[cod]

# Database & Local Files
*.db
*.sqlite3
.env

# Logs & OS artifacts
*.log
.DS_Store
```

.env

Create a .env file in the project root to store system configuration and API credentials:

```
PROJECT_NAME="Job Automation Engine"
DATABASE_URL="sqlite:///./jobs.db"
ADZUNA_APP_ID="YOUR_ACTUAL_APP_ID"
ADZUNA_APP_KEY="YOUR_ACTUAL_APP_KEY"
```

Create .env.example as a safe-to-commit template:

```
PROJECT_NAME="Job Automation Engine"
DATABASE_URL="sqlite:///./jobs.db"
ADZUNA_APP_ID="your_app_id_here"
ADZUNA_APP_KEY="your_app_key_here"
```

Step 4: Write Modular Code

1. System Settings (app/core/config.py)

Reads variables from .env cleanly using Pydantic:

```
from pydantic_settings import BaseSettings, SettingsConfigDict

class Settings(BaseSettings):
    PROJECT_NAME: str = "Job Automation Engine"
    DATABASE_URL: str = "sqlite:///./jobs.db"
    ADZUNA_APP_ID: str
    ADZUNA_APP_KEY: str

    model_config = SettingsConfigDict(env_file=".env", extra="ignore")

settings = Settings()
```

2. Database Engine (app/core/database.py)

```
from sqlmodel import SQLModel, create_engine, Session
from app.core.config import settings
```

```
# SQLite requires specific threading args for FastAPI async calls
connect_args = {"check_same_thread": False} if settings.DATABASE_URL.startswith("sqlite") else {}
engine = create_engine(settings.DATABASE_URL, connect_args=connect_args)

def init_db():
    SQLAlchemyModel.metadata.create_all(engine)

def get_session():
    with Session(engine) as session:
        yield session
```

3. Database Model (app/models/job.py)

```
from typing import Optional
from sqlmodel import SQLAlchemyModel, Field
from datetime import datetime

class JobListing(SQLAlchemyModel, table=True):
    id: Optional[int] = Field(default=None, primary_key=True)
    external_id: str = Field(unique=True, index=True)
    title: str
    company: str
    location: str
    description: str
    url: str
    source_api: str = "Adzuna"
    created_at: datetime = Field(default_factory=datetime.utcnow)
```

4. External Service Client (app/services/adzuna_client.py)

```
import httpx
from app.core.config import settings

class AdzunaClient:
    BASE_URL = "https://api.adzuna.com/v1/api/jobs/us/search/1"

    @classmethod
    async def fetch_jobs(cls, query: str = "python developer", location: str = "remote", results_per_page: int
        params = {
            "app_id": settings.ADZUNA_APP_ID,
            "app_key": settings.ADZUNA_APP_KEY,
            "results_per_page": results_per_page,
            "what": query,
            "where": location,
            "content-type": "application/json"
        }

        async with httpx.AsyncClient() as client:
            response = await client.get(cls.BASE_URL, params=params)
            response.raise_for_status()
            data = response.json()
            return data.get("results", [])
```

5. API Router Endpoint (app/api/v1/endpoints/jobs.py)

```
from fastapi import APIRouter, Depends, HTTPException, Query
from sqlmodel import Session, select
from app.core.database import get_session
from app.models.job import JobListing
from app.services.adzuna_client import AdzunaClient

router = APIRouter(prefix="/jobs", tags=["Jobs"])

@router.post("/fetch")
async def fetch_and_store_jobs(
    query: str = Query(default="python developer"),
    location: str = Query(default="remote"),
    session: Session = Depends(get_session)
):
    try:
        raw_jobs = await AdzunaClient.fetch_jobs(query=query, location=location)
    except Exception as e:
        raise HTTPException(status_code=500, detail=f"Failed to query upstream API: {str(e)}")

    added_count = 0
    for item in raw_jobs:
```

```

    ext_id = str(item.get("id"))

    # Deduplication check
    existing = session.exec(select(JobListing).where(JobListing.external_id == ext_id)).first()
    if not existing:
        job = JobListing(
            external_id=ext_id,
            title=item.get("title", "N/A"),
            company=item.get("company", {}).get("display_name", "Unknown"),
            location=item.get("location", {}).get("display_name", "Remote"),
            description=item.get("description", ""),
            url=item.get("redirect_url", "")
        )
        session.add(job)
        added_count += 1

    session.commit()
    return {"status": "success", "new_jobs_added": added_count}

@router.get("/", response_model=list[JobListing])
def get_jobs(session: Session = Depends(get_session)):
    return session.exec(select(JobListing)).all()

```

6. Application Root (app/main.py)

```

from contextlib import asynccontextmanager
from fastapi import FastAPI
from app.core.config import settings
from app.core.database import init_db
from app.api.v1.endpoints.jobs import router as jobs_router

@asynccontextmanager
async def lifespan(app: FastAPI):
    # Initialize DB tables on startup
    init_db()
    yield

app = FastAPI(
    title=settings.PROJECT_NAME,
    lifespan=lifespan
)

# Register endpoints under /api/v1
app.include_router(jobs_router, prefix="/api/v1")

```

Step 5: Test Your Application

Run Uvicorn, explicitly pointing to your modular package (app.main:app):

```
uvicorn app.main:app --reload
```

- **Verification:** Navigate to http://127.0.0.1:8000/docs in your browser. You will see the Interactive OpenAPI interface listing POST /api/v1/jobs/fetch and GET /api/v1/jobs/.
- **Test Endpoint:** Click **POST /api/v1/jobs/fetch** -> **Try it out** -> **Execute**. The application will query Adzuna, eliminate duplicates, and return the record count inserted into your local jobs.db database.

Sources

1. [FastAPI Project Structure: Production Guide 2026 - Zestminds](#)

Response: The {"detail": "Not Found"} response occurs because FastAPI does not automatically serve a root route (/) unless one is defined, and the endpoints in your project are registered under specific path prefixes.

To view your interactive OpenAPI documentation or access your endpoints:

1. Access the OpenAPI Documentation

Go to the interactive API docs route in your browser:

```
http://127.0.0.1:8000/docs
```

Or access the alternative ReDoc view:

`http://127.0.0.1:8000/redoc`

2. Access the Registered Endpoints

Based on the code structure, your endpoints are mounted under the `/api/v1` prefix:

- **Get stored jobs:** `http://127.0.0.1:8000/api/v1/jobs/`
- **Fetch new jobs (POST request):** `http://127.0.0.1:8000/api/v1/jobs/fetch`

3. (Optional) Add a Root Route (/)

If you want visiting `http://127.0.0.1:8000/` directly to return a status message instead of a 404, add a basic root route to `app/main.py`:

```
@app.get("/")
def read_root():
    return {
        "status": "online",
        "docs": "http://127.0.0.1:8000/docs",
        "jobs_endpoint": "http://127.0.0.1:8000/api/v1/jobs/"
    }
```

Save `app/main.py` and refresh `http://127.0.0.1:8000/` in your browser.

User prompt: where to get the API keys to make this work

Response: To get the **App ID** and **App Key** required for the Adzuna API, follow these steps:

1. Register for an Adzuna Developer Account

1. Go to the [Adzuna Developer Portal](#).
2. Click on **Register** to create a free account.
3. Fill out the registration form (no credit card is required).

2. Obtain Your Credentials

1. Once logged in, navigate to your developer dashboard.
2. Create or view your default application.
3. Copy your **Application ID (app_id)** and **Application Key (app_key)**.

3. Add Keys to Your Project

Paste those values directly into the `.env` file located in the root directory of your Python project:

```
PROJECT_NAME="Job Automation Engine"
DATABASE_URL="sqlite:///./jobs.db"
ADZUNA_APP_ID="paste_your_app_id_here"
ADZUNA_APP_KEY="paste_your_app_key_here"
```

Save the `.env` file and restart your Uvicorn server (`uvicorn app.main:app --reload`). The FastAPI application will read these variables on startup and send valid queries to Adzuna.

User prompt: what can I do now that the API keys are in

Response: Now that your API credentials are saved in `.env`, you can verify the end-to-end data flow, populate your database, and begin adding the core features of your automation engine.

Step 1: Test Live Ingestion

1. **Start the server** in your terminal:

```
uvicorn app.main:app --reload
```

2. **Open the interactive docs:** Navigate to `http://127.0.0.1:8000/docs` in your browser.

3. Execute a fetch request:

- Expand the `POST /api/v1/jobs/fetch` endpoint.
- Click **Try it out**.
- Enter test parameters (e.g., `query: software engineer, location: remote`).
- Click **Execute**.

4. Verify the output:

- The response should return `{"status": "success", "new_jobs_added": <number>}`.
- Expand `GET /api/v1/jobs/` and click **Execute**. You should see the raw job listings pulled directly from Adzuna and stored in your local `jobs.db` SQLite database.

Step 2: Next System Capabilities to Build

With data ingestion working, you can expand your layered backend to handle profile channels, resume matching, and job tracking:

1. Define Candidate Profiles & Contact Channels

Create a data model in `app/models/profile.py` to store your contact channels (e.g., primary email, alternate emails, Google Voice numbers). This allows the system to map specific contact methods to different job types or application variants.

2. Implement the AI Resume Matching & Tailoring Engine

Create a service module (`app/services/resume_tailor.py`) that uses an LLM API (such as OpenAI or Anthropic) to:

- Compare a job description stored in your database against a master list of your skills.
- Select relevant bullet points and output a customized, ATS-friendly resume structure.
- Render the output into a downloadable PDF format (using libraries like `WeasyPrint` or `ReportLab`).

3. Build a Background Task Scheduler

Instead of manually clicking `/fetch` in the docs, configure a task queue or scheduler (such as **APScheduler** or **Celery** with Redis) to automatically run job searches at set intervals (e.g., every 6 hours) and populate new matches in the background.

4. Design the Application Pipeline Dashboard

Create a frontend interface (or additional API routes) to manage your pipeline:

- **Kanban Statuses:** `Discovered`, `Resume Generated`, `Submitted`, `Interviewing`.
- **Channel Logs:** Track which email/phone number was assigned to which application.

User prompt: Go back to this

Response: The `{"detail": "Method Not Allowed"}` error (HTTP 405) occurs because browsers send an HTTP **GET** request whenever you type a URL into the address bar and press Enter. However, in your code structure, the `/api/v1/jobs/fetch` route was defined as an HTTP **POST** endpoint.

Option 1: Execute via Interactive Docs (Recommended)

Because `/fetch` triggers a database update, keeping it as a `POST` method follows standard REST API practices.

1. Open `[http://127.0.0.1:8000/docs](http://127.0.0.1:8000/docs)` in your browser.
2. Click on `POST /api/v1/jobs/fetch`.
3. Click **Try it out**.
4. Set your parameters (`query: software engineer`) and click **Execute**.

Option 2: Change Endpoint to `@router.get`

If you want to trigger the fetch directly by typing the URL into your browser address bar:

1. Open `app/api/v1/endpoints/jobs.py`.
2. Change the route decorator from `@router.post` to `@router.get`:

```
# Change this line:
# @router.post("/fetch")
```

```
# To this:
@router.get("/fetch")
async def fetch_and_store_jobs(
    query: str = Query(default="python developer"),
    location: str = Query(default="remote"),
    session: Session = Depends(get_session)
):
    ...
```

3. Save the file.

4. Refresh [`http://127.0.0.1:8000/api/v1/jobs/fetch?query=software%20engineer`] (`http://127.0.0.1:8000/api/v1/jobs/fetch?query=software%20engineer`) in your browser.

User prompt: [{"title": "CDL-A Shuttle Driver", "external_id": "5825159645", "location": "Dallas, Luzerne County", "url": "https://www.adzuna.com/land/ad/5825159645?se=VB9Erip8Rgll_dq_IL75Q&utm_medium=api&utm_source=94306f04&v=E3A17FC2C328E6072E56A056DC475A6364DC662E", "created_at": "2009-06T05:59:07.376587", "id": "1", "company": "Core-Mark", "description": "Job Description: CDL - A Shuttle Driver - Home Daily Location : 100 West End Rd. Wilkes-Barre, PA 18703 Schedule : Sunday - Thursday | Dispatch 11:00 PM Why Drivers Choose Us Home Daily Consistent routes & hours Day 1 Health & Wellness Benefits 401(k) with company match stock purchase plan Paid time off holidays Education Assistance & growth opportunities What You'll Do Transporting empty and loaded trailers between designated locations Performing trailer hook/unhook procedures Verifying loads ...", "source_api": "Adzuna"}, {"title": "Psychologist- Competitive Pay - Escondido, CA", "external_id": "5868601094", "location": "Escondido, San Diego County", "url": "https://www.adzuna.com/land/ad/5868601094?se=VB9Erip8Rgll_dq_IL75Q&utm_medium=api&utm_source=94306f04&v=F9C696610743C749AAC848FD707A853D9DA92C88", "created_at": "2009-06T05:59:07.388768", "id": "2", "company": "LifeStance Health", "description": "We are actively looking to hire talented Psychologists in the San Marcos area, who are passionate about patient care and committed to clinical excellence. We offer Psychologists: Sign on bonus Competitive Compensation: \$156,000 - \$178,000, compensation model based on productivity. Flexible work schedules. Generous 'above market' compensation with unlimited/uncapped earnings. Full benefits package: health, dental, vision, life, 401k (with match), paid parental leave, EAP and more. Collegial work...", "source_api": "Adzuna"}, {"title": "Licensed Clinical Social Worker - Telehealth", "external_id": "5789684530", "location": "Springfield, Lane County", "url": "https://www.adzuna.com/land/ad/5789684530?se=VB9Erip8Rgll_dq_IL75Q&utm_medium=api&utm_source=94306f04&v=6FD4FFC627D93F115ADB9387C475BDF59C0B926B", "created_at": "2009-06T05:59:07.390927", "id": "3", "company": "Headway", "description": "Licensed Clinical Social Worker (LCSW) Licensed Clinical Social Worker (LCSW) Up to \$120/hour Fully Remote or In-Person Flexible Schedule Pay & Flexibility Hourly Rate: Up to \$120 per hour. Predictable Income: Guaranteed bi-weekly payments. Total Autonomy: Set your own hours with full control over your availability. Work Your Way: Practice via telehealth, in-person, or a hybrid model. Launch or Grow Your Private Practice — Stress-Free Headway is a free platform for Licensed Social Workers that ...", "source_api": "Adzuna"}, {"title": "Respiratory Therapist RRT", "external_id": "5865730558", "location": "University Park, Dallas", "url": "https://www.adzuna.com/land/ad/5865730558?se=VB9Erip8Rgll_dq_IL75Q&utm_medium=api&utm_source=94306f04&v=BFF2FAA9E713D0D9D7981F1C3FF09F3EB629BC45", "created_at": "2009-06T05:59:07.392731", "id": "4", "company": "Medical City Dallas", "description": "Apply to Ask About Our New Rates! Are you passionate about the patient experience? At HCA Healthcare, we are committed to caring for patients with purpose and integrity. We care like family! Jump-start your career as a(an) Respiratory Therapist RRT today with Medical City Dallas. Benefits Medical City Dallas offers a total rewards package that supports the health, life, career and retirement of our colleagues. The available plans and programs include: Comprehensive medical coverage that covers ...", "source_api": "Adzuna"}, {"title": "Respiratory Therapist NICU PEDS RRT", "external_id": "5867391205", "location": "Prosper, Collin County", "url": "https://www.adzuna.com/land/ad/5867391205?se=VB9Erip8Rgll_dq_IL75Q&utm_medium=api&utm_source=94306f04&v=A064A372A706D9F4A78132D9B35A84C335796978", "created_at": "2009-06T05:59:07.395346", "id": "5", "company": "Medical City Dallas", "description": "Apply to Hear About our New Rates! Our Healthcare colleagues invested over 156,000 hours volunteering in our communities. As a(an) Respiratory Therapist NICU PEDS RRT with Medical City Dallas you can be a part of an organization that is devoted to giving back! Job Summary and Qualifications It is the responsibility of the staff respiratory therapist to demonstrate proficiency in all modalities of Respiratory Care with additional competency in NICU and Pediatric patient care. The Respiratory Therapist ...", "source_api": "Adzuna"}, {"title": "Nursing Assistant", "external_id": "5864447568", "location": "Grafton, Taylor County", "url": "https://www.adzuna.com/land/ad/5864447568?se=VB9Erip8Rgll_dq_IL75Q&utm_medium=api&utm_source=94306f04&v=B3A845E6B5903901AD2382415BB101DCA3E156FE", "created_at": "2009-06T05:59:07.397215", "id": "6", "company": "Encompass Health Rehabilitation Hospital of Morgantown", "description": "Nursing Assistant Career Opportunity (RNT) Elevate Your Nursing Assistant Role at Encompass Health Are you in search of a fulfilling career rooted in your community and heart? At Encompass Health, we extend a familial embrace from the outset, inviting you to be an influential presence in people's lives within your community. As a Nursing Assistant, your role extends beyond support, allowing you to utilize your specialized skills to provide compassionate, personalized care, and assist RNs in gui...", "source_api": "Adzuna"}, {"title": "Retail Reset Merchandiser", "external_id": "5868800668", "location": "East Muncy, Lycoming County", "url": "https://www.adzuna.com/land/ad/5868800668?se=VB9Erip8Rgll_dq_IL75Q&utm_medium=api&utm_source=94306f04&v=B2C903C5E1CAA6AB1829592F78F4A6E721C9C13C", "created_at": "2009-06T05:59:07.399553", "id": "7", "company": "SAS Retail Services", "description": "We want you to help us shape the future of shopping experiences and deliver on our purpose of connecting people with the products and experiences that enrich their lives. Joining SAS Retail Services, an Advantage Solutions company, means joining a network of 65,000 teammates serving 4,000 brands and retail customers across 40 countries. All the while, being provided the opportunities, support, and enrichment you need to grow your career. Do you like staying active, working

independently, and se...","source_api":"Adzuna"},{"title":"Staff Development Coordinator RN","external_id":"5801521856","location":"Albany, Delaware County","url":"https://www.adzuna.com/land/ad/5801521856?se=VB9Y9Erip8RGll_dq_IL75Q&utm_medium=api&utm_source=94306f04&v=6ECAFA064214CA93604B96A30B65B2FF39BCD6D3","created_at":"2019-06T05:59:07.401940","id":8,"company":"Signature HealthCARE of Muncie","description":"About Us : Welcome to Signature Healthcare of Muncie, it is a 185-bed facility. It is our mission as a family-based organization to revolutionize the long-term care industry through a culture of patient centered healthcare services, personalized spirituality, and real quality of life initiatives. The vision, to radically change the landscape of long-term care forever. Through education and empowerment, we earn the trust of every patient, family member, and the community we serve. About Signatu...","source_api":"Adzuna"},{"title":"CDL Class B Driver (Kaukauna, WI) Signing Bonus \$4,500, Monday - Friday, and Great Benefits","external_id":"5810640551","location":"Luxemburg, Kewaunee County","url":"https://www.adzuna.com/land/ad/5810640551?se=VB9Y9Erip8RGll_dq_IL75Q&utm_medium=api&utm_source=94306f04&v=D6B75DB83B3DB5DA6300D0BCE04FC6112F9267CD","created_at":"2019-06T05:59:07.404499","id":9,"company":"Orica","description":"About Orica At Orica, it's the power of our people that leads change and shapes our futures. Every day, all around the world, our people help mobilize vital resources essential to progress. Established in 1874, we have grown to become the world leader in mining and civil blasting with a diverse team of more than 13,000 across the world. From the production and supply of explosives, blasting systems, mining chemicals, and geotechnical monitoring to our cutting-edge digital solutions and comprehe...","source_api":"Adzuna"},{"title":"Project Manager – Data Science & AI Engineering","external_id":"5867162566","location":"Boston, Suffolk County","url":"https://www.adzuna.com/land/ad/5867162566?se=VB9Y9Erip8RGll_dq_IL75Q&utm_medium=api&utm_source=94306f04&v=F3E9C9C15B9E947B53F2B2C1F517163F68CC2071","created_at":"2026-09-06T05:59:07.407012","id":10,"company":"MassMutual","description":"The Opportunity At MassMutual , we're more than life insurance – we're a leading financial services company committed to helping people secure their future and protect what matters most. As a Project Manager in Data Science & AI Engineering , you'll lead the delivery of AI and data-driven solutions by managing scope, timelines, and stakeholder expectations across multiple, concurrent projects. You'll partner closely with data scientists, AI engineers, and product managers to ensure projects run...","source_api":"Adzuna"]}]

Response: Your API integration, database schema, and JSON serializer are working as expected.

What Is Happening

The Adzuna API returned default nationwide listings because no default search filters were provided in the POST request payload or query parameters. The FastAPI endpoint processed the JSON response from Adzuna and wrote all 10 entries directly into your local jobs.db SQLite database.

Step 1: Target Specific Tech & Remote Roles

To pull relevant developer jobs rather than random nationwide listings, pass explicit query and location parameters when triggering the /fetch endpoint.

Via Curl:

```
curl -X 'POST' \
  'http://127.0.0.1:8000/api/v1/jobs/fetch?query=python%20developer&location=remote' \
  -H 'accept: application/json'
```

Via Interactive Docs (http://127.0.0.1:8000/docs):

1. Expand **POST /api/v1/jobs/fetch**.
2. Click **Try it out**.
3. Set query to python developer (or software engineer).
4. Set location to remote (or your preferred city/state).
5. Click **Execute**.

Step 2: Clear Non-Tech Jobs from SQLite (Optional)

To clear out those initial test entries and start with a clean database, run a quick one-liner in your terminal using Python:

```
python3 -c "from app.core.database import engine; from sqlmodel import Session, select; from app.models.job im
```

User prompt: Delivered-To: danielblaisero@gmail.com Received: by 2002:a05:6a06:4484:b0:7b6:5b68:d8ab with SMTP id gj4csp4983178pib; Sat, 5 Sep 2026 13:38:45 -0700 (PDT) X-Received: by 2002:a05:622a:986:b0:52e:f66:f1a4 with SMTP id d75a77b69052e-530549b633amr146065881cf.38.1788640725437; Sat, 05 Sep 2026 13:38:45 -0700 (PDT) ARC-Seal: i=1; a=rsa-sha256; t=1788640725; cv=none; d=google.com; s=arc-20260327; b=TH+hXl4Z/3THlkukviFLDntyBrJgPqk56y4YfWgBv/U1fkJvsurMcXelEgP+jTJaZ mOmgoNASm2rjvOLW2kL82Rl9BtgPih3DkJDBxz7UKfd2PBL SJCKg+B5duobyh22sJoin Zxq16UH3EXNpAL7hjSmG8qfVW8kELOeWL14Gu2mk4glA+JOtL9/8765i1J1IDC3RnQK6 Xt+BFrvNHj6PIUIE+ff9Vt20hGv6OAOPxDa3agKl9aqPZaXIG1jufDVuOOenEJoaAWk SyUxSAMwjEd1FJ2WcwokDQe8wwPdEOmyoCRdPgJZLbLsHmlB8AyDhL3fN5S5AvBdJESh B+JA== ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=feedback-id:message-id:content-transfer-encoding:subject:date:to:from:mime-version:dkim-signature:dkim-signature; bh=6szbrKQBPrnGOMjghtPhpL4snBzNTVYfvJcR51OOyKE=;

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010001a0734b8818-6e9d2266-fcb8-435a-838e-1672f7ada7b0-000000@amazonses.com designates 54.240.48.97 as permitted sender)
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sp=QUARANTINE dis=NONE) header.from=ingomoney.com Return-Path: <010001a0734b8818-6e9d2266-fcb8-435a-838e-1672f7ada7b0-
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mx.google.com with ESMTPS id d75a77b69052e-53054435209si54839651cf.174.2026.09.05.13.38.45 for <danielblaiserothers@gmail.com>
(version=TLS1_3 cipher=TLS_AES_128_GCM_SHA256 bits=128/128); Sat, 05 Sep 2026 13:38:45 -0700 (PDT) Received-SPF: pass (google.com:
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sender) client-ip=54.240.48.97; Authentication-Results: mx.google.com; dkim=pass header.i=@ingomoney.com
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435a-838e-1672f7ada7b0-000000@amazonses.com designates 54.240.48.97 as permitted sender) smtp.mailfrom=010001a0734b8818-
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KhA72rCqLM8gJBD0hQIKJHh5GD+VcypP3/c8QS7zlmPLvLh9UR71zZcC/WFMHtnN4 N/u4H1tn3w== DKIM-Signature: v=1; a=rsa-sha256;
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Version: 1.0 From: info@ingomoney.com To: danielblaiserothers@gmail.com Date: Sat, 5 Sep 2026 20:38:45 +0000 Subject: Cash a Check - Check
Transaction to your account declined Content-Type: text/html; charset=us-ascii Content-Transfer-Encoding: quoted-printable Message-ID:
<010001a0734b8818-6e9d2266-fcb8-435a-838e-1672f7ada7b0-000000@email.amazonses.com> X-OriginalArrivalTime: 05 Sep 2026
20:38:44.0729 (UTC) FILETIME=[8B86DE90:01DD3D76] Feedback-ID: ::1.us-east-
1.UsY/emuqbCJ4W+cumPx/Hxc2sdA2fFw51MMCzI+Owo=:AmazonSES X-SES-Outgoing: 2026.09.05-54.240.48.97 <!DOCTYPE html PUBLIC
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family: Oxygen, Arial, sans-serif; box-sizing: border-box" = xmlns=3D"http://www.w3.org/1999/xhtml"><head></head><body
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100%; font-size: 15px; line-height: 22px; c= olor: #4d4f53; background-color: #E4EDE5"> <img alt=3D"" src=3D"http= s://vcmklc9r.r.us-
east-1.amazonaws.com/010001a0734b8818-6e9d2266-fcb8-435a= -838e-1672f7ada7b0-000000/XP3OlFgSnnSGK01Lvw7QjRCNkFA=3D473"
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padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: borde= r-box"> <tr align=3D"left" style=3D"margin: 0; padding: 0; font-family: Oxyg= n,
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600px" valign=3D"to= p"> <!-- shared header start --> <table align=3D"center" class=3D"full-wi= dth-container" style=3D"margin: 0; padding: 0;
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=3D"77"src=3D"https://ingo-transactional-emails.s3.amazonaws.com/PayPal+-+P= repaid/logo3.0.png" style=3D"margin: 0px; padding: 0px; font-
family: Oxygen, = Arial, sans-serif; box-sizing: border-box; outline: none; text-decoration: = none; color: rgb(35, 173, 75); font-size:
24px;"title=3D"card" width=3D"70" = " /> <!-- logo end --> </td><!-- card number start --> <td align=3D"right" = " class=3D"cardnum"
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align: right; padding-right: 2%; font-size: 12px; line-height: 18px; = padding-bottom: 20px" valign=3D"bottom"><img border=3D"0"
height=3D"36.8= 64" style=3D"padding-right: 20px"src=3D"https://ingo-transactional-emails.s= 3.amazonaws.com/PayPal+-+Prepaid/PayPal-
Logo-FINAL.png" width=3D"145" /> <= /td><!-- card number end --> <td align=3D"right" class=3D"cardnum" height= =3D"60" style=3D"margin:
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size: 12px; line-height: 18px; padding-bottom: 20px" v= align=3D"bottom"><img border=3D"0" height=3D"25"style=3D"border-left-width= 2px;
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wordmark.png"width=3D"145" /> </td></tr></tbody></table><!-- shared header end --><!-- body content start --></td></tr></tbody></table>

align=3D"left" style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left"><td align=3D"left" class=3D"fixed-width-container container-gutter" style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left; padding-left: 2%; padding-right: 2%; width: 600px" valign=3D"top"><table cellpadding=3D"0" cellspacing=3D"0" class=3D"main" style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left; margin-top: 8px; background-color: #FFF; border-collapse: separate; border-radius: 8px; border: solid 1px #E4EDE5; box-shadow: 0px 4px 10px 1px #E4EDE5; -webkit-box-shadow: 0px 4px 10px 1px #E4EDE5; -moz-box-shadow: 0px 4px 10px 1px #E4EDE5"><tbody style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left"><td align=3D"left" style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left; padding-top: 35px; padding-bottom: 35px" valign=3D"top"><table cellpadding=3D"0" cellspacing=3D"0" style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left"><tbody style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left; padding-top: 35px; padding-bottom: 35px"><tr align=3D"left" style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left; padding-top: 35px; padding-bottom: 35px"><td align=3D"left" class=3D"content-gutter" style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left; padding-left: 6%; padding-right: 6%" valign=3D"top"><h1 class=3D"greeting" style=3D"margin: 0; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; color: #23ad4b; font-size: 20px; line-height: 27px; font-weight: normal; margin-bottom: 20px">Transaction Notice</h1><h1 class=3D"greeting" style=3D"margin: 0; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; color: #23ad4b; font-size: 20px; line-height: 27px; font-weight: normal; margin-bottom: 20px">Your check transaction is not approved</h1><!-- celebratory message start --><!-- celebratory message end --><p style=3D"margin: 0px; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; margin-bottom: 20px; color: #4d4f53; font-weight: normal">Dear Daniel Blaise Rogers,</p><p style=3D"margin: 0px; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; margin-bottom: 20px; color: #4d4f53; font-weight: normal">Unfortunately, we are unable to approve your check transaction with Transaction ID b2bd439e-ebc2-4166-8cc2-a8e879140ae3, dated 9/5/202= 6. Your check will not be processed and your account will not be funded.</p><p style=3D"margin: 0px; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; margin-bottom: 20px; color: #4d4f53; font-weight: normal">Your check remains valid and may be deposited or cashed elsewhere at a later time.</p><p style=3D"margin: 0px; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; margin-bottom: 20px; color: #4d4f53; font-weight: normal">Checks may not be approved for a variety of reasons, including:</p><ul style=3D"margin: 0px; padding: 0px; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; color: #23ad4b; font-weight: normal;"><li style=3D"margin: 0px; padding: 0px; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; color: #23ad4b; font-weight: normal;">Check was not endorsed<li style=3D"margin: 0px; padding: 0px; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; color: #23ad4b; font-weight: normal;">Check images are illegible<li style=3D"margin: 0px; padding: 0px; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; color: #23ad4b; font-weight: normal;">Check was not written to you as your name appears on your account<li style=3D"margin: 0px; padding: 0px; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; color: #23ad4b; font-weight: normal;">A properly voided image of the check was not received<p style=3D"margin: 0px; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; margin-bottom: 20px; color: #4d4f53; font-weight: normal">You may retain this email for your records. You can also access your check transaction at any time as follows:</p><ul style=3D"margin: 0px; padding: 0px; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; color: #23ad4b; font-weight: normal;"><li style=3D"margin: 0px; padding: 0px; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; color: #23ad4b; font-weight: normal;">If you were using PayPal, open the app and launch "Cash a Check" from the button on the home screen, then select the menu icon (three dots) in the top right and view "My History". Choose the check transaction you wish to complete.<li style=3D"margin: 0px; padding: 0px; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; color: #23ad4b; font-weight: normal;">If you were using Venmo, open the app, go to the Menu, tap "Manage Balance" and select "Cash a check".<p style=3D"margin: 0px; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; margin-bottom: 10px; color: #4d4f53; font-weight: normal">If you have any questions or concerns about your transaction, please contact Ingo Money at help@ingomoney.com or call (229) 276-3955.</p><p style=3D"margin: 0px; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; margin-bottom: 10px; color: #4d4f53; font-weight: normal">Thank you. We hope that you will try us again in the future.</p><p style=3D"margin: 0px; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; margin-bottom: 10px; color: #4d4f53; font-weight: normal">Ingo Money Cash a Check Customer Service</p></td></tr></tbody></table><!-- secondary message container end --></td></tr></tbody></table><!-- footer container start --><table class=3D"footer" style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left"><tbody style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left"><tr align=3D"left" style=3D"margin: 0; padding: 0; font-family: Oxygen, Arial, sans-serif; box-sizing: border-box; vertical-align: top; text-align: left; padding-top: 40px" valign=3D"top"><td align=3D"left" style=3D"margin: 0; padding: 0; font-family: Oxygen, Helvetica, Arial, sans-serif; box-sizing: border-box; margin-bottom: 10px; color: #4d4f53; font-weight: normal; line-height: 16px; font-size: 11px">The Cash a Check feature is a service provided by Sunrise Banks, N.A. - Member FDIC and Ingo Money, Inc., subject to the Sunrise Banks and Ingo Money service

```
--> </td></tr> </tbody> </table><!-- footer container= end --> <!-- body content end --></td> </tr> </tbody></table> </center> = </td></tr>
</tbody> </table></div><!-- shared layout end -->
```

Response: Based on the email notification, here is the summary of your check cashing transaction status:

- **Status:** Declined (Not Approved)
- **Transaction ID:** b2bd439e-ebc2-4166-8cc2-a8e879140ae3
- **Date:** September 5, 2026
- **Service Provider:** Ingo Money (via PayPal / Venmo Cash a Check)

Common Reasons for Declining

Check transactions through Ingo Money are generally not approved due to one of the following issues:

- **Endorsement:** The check was missing a signature on the back.
- **Image Quality:** The submitted photos of the check were blurry or illegible.
- **Name Mismatch:** The payee name on the check did not match your account name exactly.
- **Voided Image:** A required image showing a properly voided check was not received.

Next Steps & Resolution

1. **Check Validity:** Your physical check remains valid and has not been processed or canceled; you can still deposit or cash it elsewhere.
2. **Review in App:**
 - **PayPal:** Open the app, launch "Cash a Check," tap the menu icon (. . .) in the top right, and select **My History**.
 - **Venmo:** Open the app, go to **Menu > Manage Balance > Cash a check**.
3. **Support:** If you need further clarification, contact Ingo Money Customer Service directly:
 - **Email:** help@ingomoney.com
 - **Phone:** (229) 276-3955

User prompt: AHPWFDEE UWIKUDFC VR44XDRK N5NADGSD 3I57Y5BB 5QDUGSR4 XRDF7K3E ZNOWFEUT

Response: The backup codes in your text list and screenshot match. Here is the list formatted for quick copying or saving to a password manager:

```
AHPWFDEE
UWIKUDFC
VR44XDRK
N5NADGSD
3I57Y5BB
5QDUGSR4
XRDF7K3E
ZNOWFEUT
```

Important Security Reminders

- **Usage:** Each code can only be used once to log in if you lose access to your primary two-factor authentication device.
- **Storage:** Store these codes in a secure location, such as a password manager or an encrypted file.
- **Regeneration:** If you generate a new set of backup codes in your Squarespace account settings later, this current set will automatically become invalid.

User prompt: Can you put all this output into as few commands so I can copy and paste when this Google account wipes the Linux environment on this computer: NOTICE: To simplify system architecture and maintenance, Crostini has switched by default to a containerless design for new environments starting in ChromeOS version 143 and newer. If you experience unexpected issues with the new design, please report them using the instructions available at <https://www.chromium.org/chromium-os/developer-library/guides/bugs/platform-public-tracker/>. If you would like to revert to the previous system architecture, you may visit <chrome://flags#containerless-crostini> in your Chrome browser and set the flag to "Disabled", then restart your device, and un- and reinstall Crostini. PSA: In order to mitigate the Januscape kernel exploit (see: <https://github.com/V4bel/Januscape>), nested virtualization has been disabled for host kernel versions 5.10 and 5.15. To check your host

kernel version, you can navigate to `chrome://system`, find the entry named 'uname', and if it looks something like 'Linux localhost 5.10.XXXX' or 'Linux localhost 5.15.XXX' nested virtualization will be disabled (this message will be repeated 4 more times). (to silence this message, run the following command): `echo 10 > "/home/date09062026/.local/share/baguette-motd" date09062026@penguin:~$ sudo apt update && sudo apt upgrade -y` Hit:1 <https://deb.debian.org/debian trixie InRelease> Ign:2 <https://storage.googleapis.com/cros-packages-staging/153 trixie InRelease> Hit:3 <https://storage.googleapis.com/cros-packages-staging/153 trixie Release> Get:5 <https://deb.debian.org/debian trixie/main amd64 Components> [5175 kB] Fetched 5175 kB in 3s (1699 kB/s) All packages are up to date. Summary: Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0 date09062026@penguin:~\$ sudo apt install -y build-essential git curl wget unzip ca-certificates gnupg curl is already the newest version (8.14-1-2+deb13u4). wget is already the newest version (1.25.0-2). ca-certificates is already the newest version (20250419). Installing: build-essential git gnupg unzip Installing dependencies: binutils gpg-agent libfile-fcntllock-perl binutils-common gpg-wks-client libgcc-14-dev binutils-x86-64-linux-gnu gpgsm libgprofng0 bzip2 gpgv libhwasa0 dirmngr libalgorithm-diff-perl libitm1 dpkg-dev libalgorithm-diff-xs-perl liblsan0 fakeroot libalgorithm-merge-perl libquadmath0 g++ libasan8 libstdc++14 libbinutils libubsan1 gcc libcc1-0 linux-libc-dev gcc-14 libcrypt-dev make gcc-14-x86-64-linux-gnu libctf-nobfd0 manpages gcc-x86-64-linux-gnu libctf0 manpages-dev git-man libdpkg-perl patch gnupg-l10n liberror-perl pinentry-curses gnupg-utils libfakeroot rpcsvc-proto Suggested packages: binutils-doc autoconf git-email libc-devtools gprofng-gui automake git-gui glibc-doc binutils-gold libtool gitk bzip2-doc flex gitweb libstdc++14-doc pinentry-gnome3 bison git-cvs make-doc tor gdb git-mediawiki man-browser debian-keyring gcc-doc git-svn ed debian-tag2upload-keyring gcc-14-multilib gpg-wks-server diffutils-doc g++-multilib gcc-14-locales parcimonie pinentry-doc g++-14-multilib gdb-x86-64-linux-gnu xloadimage zip gcc-14-doc gettext-base sdaemon gcc-multilib git-doc tpm2daemon Summary: Upgrading: 0, Installing: 58, Removing: 0, Not Upgrading: 0 Download size: 79.2 MB Space needed: 313 MB / 14.5 GB available Get:1 <https://deb.debian.org/debian trixie/main amd64 bzip2 amd64 1.0.8-6> [40.5 kB] Get:2 <https://deb.debian.org/debian trixie/main amd64 manpages all 6.9.1-1> [1393 kB] Get:3 <https://deb.debian.org/debian trixie/main amd64 libstdc++14 amd64 14.2.0-19> [26.0 kB] Get:4 <https://deb.debian.org/debian trixie/main amd64 libbinutils amd64 2.44-3> [2509 kB] Get:5 <https://deb.debian.org/debian trixie/main amd64 libbinutils amd64 2.44-3> [534 kB] Get:6 <https://deb.debian.org/debian trixie/main amd64 libgprofng0 amd64 2.44-3> [808 kB] Get:7 <https://deb.debian.org/debian trixie/main amd64 libctf-nobfd0 amd64 2.44-3> [156 kB] Get:8 <https://deb.debian.org/debian trixie/main amd64 libctf0 amd64 2.44-3> [88.6 kB] Get:9 <https://deb.debian.org/debian trixie/main amd64 binutils-x86-64-linux-gnu amd64 2.44-3> [1014 kB] Get:10 <https://deb.debian.org/debian trixie/main amd64 binutils amd64 2.44-3> [265 kB] Get:11 <https://deb.debian.org/debian trixie/main amd64 libc-dev-bin amd64 2.41-12+deb13u3> [59.8 kB] Get:12 <https://deb.debian.org/debian trixie/main amd64 linux-libc-dev all 6.12.94-1> [2848 kB] Get:13 <https://deb.debian.org/debian trixie/main amd64 libcrypt-dev amd64 1:4.4.38-1> [119 kB] Get:14 <https://deb.debian.org/debian trixie/main amd64 rpcsvc-proto amd64 1.4.3-1> [63.3 kB] Get:15 <https://deb.debian.org/debian trixie/main amd64 libc6-dev amd64 2.41-12+deb13u3> [1992 kB] Get:16 <https://deb.debian.org/debian trixie/main amd64 libcc1-0 amd64 14.2.0-19> [42.8 kB] Get:17 <https://deb.debian.org/debian trixie/main amd64 libitm1 amd64 14.2.0-19> [26.0 kB] Get:18 <https://deb.debian.org/debian trixie/main amd64 libasan8 amd64 14.2.0-19> [2725 kB] Get:19 <https://deb.debian.org/debian trixie/main amd64 liblsan0 amd64 14.2.0-19> [1204 kB] Get:20 <https://deb.debian.org/debian trixie/main amd64 libstdc++14 amd64 14.2.0-19> [2460 kB] Get:21 <https://deb.debian.org/debian trixie/main amd64 libubsan1 amd64 14.2.0-19> [1074 kB] Get:22 <https://deb.debian.org/debian trixie/main amd64 libhwasa0 amd64 14.2.0-19> [1488 kB] Get:23 <https://deb.debian.org/debian trixie/main amd64 libquadmath0 amd64 14.2.0-19> [145 kB] Get:24 <https://deb.debian.org/debian trixie/main amd64 libgcc-14-dev amd64 14.2.0-19> [2672 kB] Get:25 <https://deb.debian.org/debian trixie/main amd64 gcc-14-x86-64-linux-gnu amd64 14.2.0-19> [21.4 MB] Get:26 <https://deb.debian.org/debian trixie/main amd64 gcc-14 amd64 14.2.0-19> [540 kB] Get:27 <https://deb.debian.org/debian trixie/main amd64 gcc-x86-64-linux-gnu amd64 4:14.2.0-1> [1436 B] Get:28 <https://deb.debian.org/debian trixie/main amd64 gcc amd64 4:14.2.0-1> [5136 B] Get:29 <https://deb.debian.org/debian trixie/main amd64 libstdc++14-dev amd64 14.2.0-19> [2376 kB] Get:30 <https://deb.debian.org/debian trixie/main amd64 g++-14-x86-64-linux-gnu amd64 14.2.0-19> [12.1 MB] Get:31 <https://deb.debian.org/debian trixie/main amd64 g++ amd64 14.2.0-19> [22.5 kB] Get:32 <https://deb.debian.org/debian trixie/main amd64 g++ amd64 4:14.2.0-1> [1200 B] Get:33 <https://deb.debian.org/debian trixie/main amd64 g++ amd64 4:14.2.0-1> [1344 B] Get:34 <https://deb.debian.org/debian trixie/main amd64 make amd64 4.4.1-2> [463 kB] Get:35 <https://deb.debian.org/debian trixie/main amd64 libdpkg-perl all 1.22.22> [651 kB] Get:36 <https://deb.debian.org/debian trixie/main amd64 patch amd64 2.8-2> [134 kB] Get:37 <https://deb.debian.org/debian trixie/main amd64 dpkg-dev all 1.22.22> [1337 kB] Get:38 <https://deb.debian.org/debian trixie/main amd64 build-essential amd64 12.12> [4624 B] Get:39 <https://deb.debian.org/debian trixie/main amd64 dirmngr amd64 2.4.7-21+deb13u1+b4> [384 kB] Get:40 <https://deb.debian.org/debian trixie/main amd64 libfakeroot amd64 1.37.1-1> [29.6 kB] Get:41 <https://deb.debian.org/debian trixie/main amd64 fakeroot amd64 1.37.1-1> [76.0 kB] Get:42 <https://deb.debian.org/debian trixie/main amd64 liberror-perl all 0.17030-1> [26.9 kB] Get:43 <https://deb.debian.org/debian trixie/main amd64 git-man all 1:2.47.3-0+deb13u1> [2205 kB] Get:44 <https://deb.debian.org/debian trixie/main amd64 git amd64 1:2.47.3-0+deb13u1> [8862 kB] Get:45 <https://deb.debian.org/debian trixie/main amd64 gnupg-l10n all 2.4.7-21+deb13u1> [749 kB] Get:46 <https://deb.debian.org/debian trixie/main amd64 pinentry-curses amd64 1.3.1-2> [86.4 kB] Get:47 <https://deb.debian.org/debian trixie/main amd64 gpg-agent amd64 2.4.7-21+deb13u1+b4> [271 kB] Get:48 <https://deb.debian.org/debian trixie/main amd64 gpgsm amd64 2.4.7-21+deb13u1+b4> [276 kB] Get:49 <https://deb.debian.org/debian trixie/main amd64 gnupg all 2.4.7-21+deb13u1> [417 kB] Get:50 <https://deb.debian.org/debian trixie/main amd64 gpg-wks-client amd64 2.4.7-21+deb13u1+b4> [109 kB] Get:51 <https://deb.debian.org/debian trixie/main amd64 gpgv amd64 2.4.7-21+deb13u1+b4> [241 kB] Get:52 <https://deb.debian.org/debian trixie/main amd64 libalgorithm-diff-perl all 1.201-1> [43.3 kB] Get:53 <https://deb.debian.org/debian trixie/main amd64 libalgorithm-diff-xs-perl amd64 0.04-9> [11.1 kB] Get:54 <https://deb.debian.org/debian trixie/main amd64 libalgorithm-merge-perl all 0.08-5> [11.8 kB] Get:55 <https://deb.debian.org/debian trixie/main amd64 libfile-fcntllock-perl amd64 0.22-4+b4> [34.6 kB] Get:56 <https://deb.debian.org/debian trixie/main amd64 manpages-dev all 6.9.1-1> [2122 kB] Get:57 <https://deb.debian.org/debian trixie/main amd64 unzip amd64 6.0-29> [173 kB] Get:58 <https://deb.debian.org/debian trixie/main amd64 gnupg-utils amd64 2.4.7-21+deb13u1+b4> [194 kB] Fetched 79.2 MB in 38s (2070 kB/s) Extracting templates from packages: 100% Selecting previously unselected package bzip2. (Reading database ... 35202 files and directories currently installed.) Preparing to unpack .../00-bzip2_1.0.8-6_amd64.deb ... Unpacking bzip2 (1.0.8-6) ... Selecting previously unselected package manpages. Preparing to unpack .../01-manpages_6.9.1-1_all.deb ... Unpacking manpages (6.9.1-1) ... Selecting previously unselected package libstdc++14:amd64. Preparing to unpack .../02-libstdc++14_14.2.0-19_amd64.deb ... Unpacking libstdc++14:amd64 (14.2.0-19) ... Selecting previously unselected package binutils-common:amd64. Preparing to unpack .../03-binutils-common_2.44-3_amd64.deb ... Unpacking binutils-common:amd64 (2.44-3) ... Selecting previously unselected package libbinutils:amd64. Preparing to unpack .../04-libbinutils_2.44-3_amd64.deb ... Unpacking libbinutils:amd64 (2.44-3) ... Selecting previously unselected package libgprofng0:amd64. Preparing to unpack .../05-libgprofng0_2.44-3_amd64.deb ... Unpacking libgprofng0:amd64 (2.44-3) ... Selecting previously unselected package libctf-nobfd0:amd64.

Preparing to unpack .../06-libctf-nobfd0_2.44-3_amd64.deb ... Unpacking libctf-nobfd0:amd64 (2.44-3) ... Selecting previously unselected package libctf0:amd64. Preparing to unpack .../07-libctf0_2.44-3_amd64.deb ... Unpacking libctf0:amd64 (2.44-3) ... Selecting previously unselected package binutils-x86-64-linux-gnu. Preparing to unpack .../08-binutils-x86-64-linux-gnu_2.44-3_amd64.deb ... Unpacking binutils-x86-64-linux-gnu (2.44-3) ... Selecting previously unselected package binutils. Preparing to unpack .../09-binutils_2.44-3_amd64.deb ... Unpacking binutils (2.44-3) ... Selecting previously unselected package libc-dev-bin. Preparing to unpack .../10-libc-dev-bin_2.41-12+deb13u3_amd64.deb ... Unpacking libc-dev-bin (2.41-12+deb13u3) ... Selecting previously unselected package linux-libc-dev. Preparing to unpack .../11-linux-libc-dev_6.12.94-1_all.deb ... Unpacking linux-libc-dev (6.12.94-1) ... Selecting previously unselected package libcrypt-dev:amd64. Preparing to unpack .../12-libcrypt-dev_1%3a4.4.38-1_amd64.deb ... Unpacking libcrypt-dev:amd64 (1:4.4.38-1) ... Selecting previously unselected package rpcsvc-proto. Preparing to unpack .../13-rpcsvc-proto_1.4.3-1_amd64.deb ... Unpacking rpcsvc-proto (1.4.3-1) ... Selecting previously unselected package libc6-dev:amd64. Preparing to unpack .../14-libc6-dev_2.41-12+deb13u3_amd64.deb ... Unpacking libc6-dev:amd64 (2.41-12+deb13u3) ... Selecting previously unselected package libcc1-0:amd64. Preparing to unpack .../15-libcc1-0_14.2.0-19_amd64.deb ... Unpacking libcc1-0:amd64 (14.2.0-19) ... Selecting previously unselected package libitm1:amd64. Preparing to unpack .../16-libitm1_14.2.0-19_amd64.deb ... Unpacking libitm1:amd64 (14.2.0-19) ... Selecting previously unselected package libasan8:amd64. Preparing to unpack .../17-libasan8_14.2.0-19_amd64.deb ... Unpacking libasan8:amd64 (14.2.0-19) ... Selecting previously unselected package liblsan0:amd64. Preparing to unpack .../18-liblsan0_14.2.0-19_amd64.deb ... Unpacking liblsan0:amd64 (14.2.0-19) ... Selecting previously unselected package libtsan2:amd64. Preparing to unpack .../19-libtsan2_14.2.0-19_amd64.deb ... Unpacking libtsan2:amd64 (14.2.0-19) ... Selecting previously unselected package libubsan1:amd64. Preparing to unpack .../20-libubsan1_14.2.0-19_amd64.deb ... Unpacking libubsan1:amd64 (14.2.0-19) ... Selecting previously unselected package libhwasan0:amd64. Preparing to unpack .../21-libhwasan0_14.2.0-19_amd64.deb ... Unpacking libhwasan0:amd64 (14.2.0-19) ... Selecting previously unselected package libquadmath0:amd64. Preparing to unpack .../22-libquadmath0_14.2.0-19_amd64.deb ... Unpacking libquadmath0:amd64 (14.2.0-19) ... Selecting previously unselected package libgcc-14-dev:amd64. Preparing to unpack .../23-libgcc-14-dev_14.2.0-19_amd64.deb ... Unpacking libgcc-14-dev:amd64 (14.2.0-19) ... Selecting previously unselected package gcc-14-x86-64-linux-gnu. Preparing to unpack .../24-gcc-14-x86-64-linux-gnu_14.2.0-19_amd64.deb ... Unpacking gcc-14-x86-64-linux-gnu (14.2.0-19) ... Selecting previously unselected package gcc-14. Preparing to unpack .../25-gcc-14_14.2.0-19_amd64.deb ... Unpacking gcc-14 (14.2.0-19) ... Selecting previously unselected package gcc-x86-64-linux-gnu. Preparing to unpack .../26-gcc-x86-64-linux-gnu_4%3a14.2.0-1_amd64.deb ... Unpacking gcc-x86-64-linux-gnu (4:14.2.0-1) ... Selecting previously unselected package gcc. Preparing to unpack .../27-gcc_4%3a14.2.0-1_amd64.deb ... Unpacking gcc (4:14.2.0-1) ... Selecting previously unselected package libstdc++-14-dev:amd64. Preparing to unpack .../28-libstdc++-14-dev_14.2.0-19_amd64.deb ... Unpacking libstdc++-14-dev:amd64 (14.2.0-19) ... Selecting previously unselected package g++-14-x86-64-linux-gnu. Preparing to unpack .../29-g++-14-x86-64-linux-gnu_14.2.0-19_amd64.deb ... Unpacking g++-14-x86-64-linux-gnu (14.2.0-19) ... Selecting previously unselected package g++-14. Preparing to unpack .../30-g++-14_14.2.0-19_amd64.deb ... Unpacking g++-14 (14.2.0-19) ... Selecting previously unselected package g++-x86-64-linux-gnu. Preparing to unpack .../31-g++-x86-64-linux-gnu_4%3a14.2.0-1_amd64.deb ... Unpacking g++-x86-64-linux-gnu (4:14.2.0-1) ... Selecting previously unselected package g++. Preparing to unpack .../32-g++_4%3a14.2.0-1_amd64.deb ... Unpacking g++ (4:14.2.0-1) ... Selecting previously unselected package make. Preparing to unpack .../33-make_4.4.1-2_amd64.deb ... Unpacking make (4.4.1-2) ... Selecting previously unselected package libdpkg-perl. Preparing to unpack .../34-libdpkg-perl_1.22.22_all.deb ... Unpacking libdpkg-perl (1.22.22) ... Selecting previously unselected package patch. Preparing to unpack .../35-patch_2.8-2_amd64.deb ... Unpacking patch (2.8-2) ... Selecting previously unselected package dpkg-dev. Preparing to unpack .../36-dpkg-dev_1.22.22_all.deb ... Unpacking dpkg-dev (1.22.22) ... Selecting previously unselected package build-essential. Preparing to unpack .../37-build-essential_12.12_amd64.deb ... Unpacking build-essential (12.12) ... Selecting previously unselected package dirmngr. Preparing to unpack .../38-dirmngr_2.4.7-21+deb13u1+b4_amd64.deb ... Unpacking dirmngr (2.4.7-21+deb13u1+b4) ... Selecting previously unselected package libfakeroot:amd64. Preparing to unpack .../39-libfakeroot_1.37.1.1-1_amd64.deb ... Unpacking libfakeroot:amd64 (1.37.1.1-1) ... Selecting previously unselected package fakeroot. Preparing to unpack .../40-fakeroot_1.37.1.1-1_amd64.deb ... Unpacking fakeroot (1.37.1.1-1) ... Selecting previously unselected package liberror-perl. Preparing to unpack .../41-liberror-perl_0.17030-1_all.deb ... Unpacking liberror-perl (0.17030-1) ... Selecting previously unselected package git-man. Preparing to unpack .../42-git-man_1%3a2.47.3-0+deb13u1_all.deb ... Unpacking git-man (1:2.47.3-0+deb13u1) ... Selecting previously unselected package git. Preparing to unpack .../43-git_1%3a2.47.3-0+deb13u1_amd64.deb ... Unpacking git (1:2.47.3-0+deb13u1) ... Selecting previously unselected package gnupg-l10n. Preparing to unpack .../44-gnupg-l10n_2.4.7-21+deb13u1_all.deb ... Unpacking gnupg-l10n (2.4.7-21+deb13u1) ... Selecting previously unselected package pinentry-curses. Preparing to unpack .../45-pinentry-curses_1.3.1-2_amd64.deb ... Unpacking pinentry-curses (1.3.1-2) ... Selecting previously unselected package gpg-agent. Preparing to unpack .../46-gpg-agent_2.4.7-21+deb13u1+b4_amd64.deb ... Unpacking gpg-agent (2.4.7-21+deb13u1+b4) ... Selecting previously unselected package gpgsm. Preparing to unpack .../47-gpgsm_2.4.7-21+deb13u1+b4_amd64.deb ... Unpacking gpgsm (2.4.7-21+deb13u1+b4) ... Selecting previously unselected package gnupg. Preparing to unpack .../48-gnupg_2.4.7-21+deb13u1_all.deb ... Unpacking gnupg (2.4.7-21+deb13u1) ... Selecting previously unselected package gpg-wks-client. Preparing to unpack .../49-gpg-wks-client_2.4.7-21+deb13u1+b4_amd64.deb ... Unpacking gpg-wks-client (2.4.7-21+deb13u1+b4) ... Selecting previously unselected package gpgv. Preparing to unpack .../50-gpgv_2.4.7-21+deb13u1+b4_amd64.deb ... Unpacking gpgv (2.4.7-21+deb13u1+b4) ... Selecting previously unselected package libalgorithm-diff-perl. Preparing to unpack .../51-libalgorithm-diff-perl_1.201-1_all.deb ... Unpacking libalgorithm-diff-perl (1.201-1) ... Selecting previously unselected package libalgorithm-diff-xs-perl. Preparing to unpack .../52-libalgorithm-diff-xs-perl_0.04-9_amd64.deb ... Unpacking libalgorithm-diff-xs-perl (0.04-9) ... Selecting previously unselected package libalgorithm-merge-perl. Preparing to unpack .../53-libalgorithm-merge-perl_0.08-5_all.deb ... Unpacking libalgorithm-merge-perl (0.08-5) ... Selecting previously unselected package libfile-fcntllock-perl. Preparing to unpack .../54-libfile-fcntllock-perl_0.22-4+b4_amd64.deb ... Unpacking libfile-fcntllock-perl (0.22-4+b4) ... Selecting previously unselected package manpages-dev. Preparing to unpack .../55-manpages-dev_6.9.1-1_all.deb ... Unpacking manpages-dev (6.9.1-1) ... Selecting previously unselected package unzip. Preparing to unpack .../56-unzip_6.0-29_amd64.deb ... Unpacking unzip (6.0-29) ... Selecting previously unselected package gnupg-utils. Preparing to unpack .../57-gnupg-utils_2.4.7-21+deb13u1+b4_amd64.deb ... Unpacking gnupg-utils (2.4.7-21+deb13u1+b4) ... Setting up pinentry-curses (1.3.1-2) ... Setting up gpg-agent (2.4.7-21+deb13u1+b4) ... Created symlink '/etc/systemd/user/sockets.target.wants/gpg-agent-browser.socket' → '/usr/lib/systemd/user/gpg-agent-browser.socket'. Created symlink '/etc/systemd/user/sockets.target.wants/gpg-agent-extra.socket' → '/usr/lib/systemd/user/gpg-agent-extra.socket'. Created symlink '/etc/systemd/user/sockets.target.wants/gpg-agent-ssh.socket' → '/usr/lib/systemd/user/gpg-agent-ssh.socket'. Created symlink '/etc/systemd/user/sockets.target.wants/gpg-agent.socket' → '/usr/lib/systemd/user/gpg-agent.socket'. Setting up libfile-fcntllock-perl (0.22-4+b4) ... Setting up libalgorithm-diff-perl (1.201-1) ... Setting up manpages (6.9.1-1) ... Setting up unzip (6.0-29) ... Setting up gpgsm (2.4.7-21+deb13u1+b4) ... Setting up binutils-common:amd64 (2.44-3) ... Setting up linux-libc-dev (6.12.94-1) ... Setting up libctf-nobfd0:amd64 (2.44-3) ... Setting up bzip2 (1.0.8-6) ... Setting up libsframe1:amd64 (2.44-3) ... Setting up libfakeroot:amd64 (1.37.1.1-1) ... Setting up fakeroot (1.37.1.1-1) ... update-alternatives: using /usr/bin/fakeroot-sysv to provide /usr/bin/fakeroot (fakeroot) in auto mode Setting up liberror-perl (0.17030-1) ... Setting up dirmngr (2.4.7-21+deb13u1+b4) ... Created symlink '/etc/systemd/user/sockets.target.wants/dirmngr.socket' → '/usr/lib/systemd/user/dirmngr.socket'. Setting up rpcsvc-proto (1.4.3-1) ... Setting up

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make (4.4.1-2) ... Setting up gnupg-l10n (2.4.7-21+deb13u1) ... Setting up libquadmath0:amd64 (14.2.0-19) ... Setting up patch (2.8-2) ... Setting up
libdpkg-perl (1.22.22) ... Setting up libubsan1:amd64 (14.2.0-19) ... Setting up libhwasa0:amd64 (14.2.0-19) ... Setting up gpgv (2.4.7-
21+deb13u1+b4) ... Setting up libcrypt-dev:amd64 (1:4.4.38-1) ... Setting up libasan8:amd64 (14.2.0-19) ... Setting up gnupg (2.4.7-21+deb13u1) ...
Setting up git-man (1:2.47.3-0+deb13u1) ... Setting up libtsan2:amd64 (14.2.0-19) ... Setting up libbinutils:amd64 (2.44-3) ... Setting up libc-dev-bin
(2.41-12+deb13u3) ... Setting up libalgorithm-diff-xs-perl (0.04-9) ... Setting up libcc1-0:amd64 (14.2.0-19) ... Setting up liblsan0:amd64 (14.2.0-19)
... Setting up libitm1:amd64 (14.2.0-19) ... Setting up libalgorithm-merge-perl (0.08-5) ... Setting up gnupg-utils (2.4.7-21+deb13u1+b4) ... Setting
up libctf0:amd64 (2.44-3) ... Setting up manpages-dev (6.9.1-1) ... Setting up gpg-wks-client (2.4.7-21+deb13u1+b4) ... Setting up
libgprofng0:amd64 (2.44-3) ... Setting up git (1:2.47.3-0+deb13u1) ... Setting up libcc-dev:amd64 (2.41-12+deb13u3) ... Setting up libgcc-14-
dev:amd64 (14.2.0-19) ... Setting up libstdc++-14-dev:amd64 (14.2.0-19) ... Setting up binutils-x86-64-linux-gnu (2.44-3) ... Setting up binutils
(2.44-3) ... Setting up dpkg-dev (1.22.22) ... Setting up gcc-14-x86-64-linux-gnu (14.2.0-19) ... Setting up gcc-x86-64-linux-gnu (4:14.2.0-1) ...
Setting up gcc-14 (14.2.0-19) ... Setting up g++-14-x86-64-linux-gnu (14.2.0-19) ... Setting up g++-x86-64-linux-gnu (4:14.2.0-1) ... Setting up
g++-14 (14.2.0-19) ... Setting up gcc (4:14.2.0-1) ... Setting up g++ (4:14.2.0-1) ... update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++
(c++) in auto mode Setting up build-essential (12.12) ... Processing triggers for libc-bin (2.41-12+deb13u3) ... date09062026@penguin:~$ curl -X
'GET' \ 'http://127.0.0.1:8000/api/v1/jobs/fetch?query=developer%20&location=remote%20' \ -H 'accept: application/json'
date09062026@penguin:~$ curl -X 'GET' \ 'http://127.0.0.1:8000/api/v1/jobs/fetch?query=%20&location=28716%20' \ -H 'accept:
application/json' {"status":"success","new_jobs_added":10}date09062026@penguin:~$ curl -X 'GET' 'http://127.0.0.1:8000/api/v1/jobs/fetch?
query=%20&location=32304%20' -H 'accept: application/json' {"status":"success","new_jobs_added":10}date09062026@penguin:~$ git add .
fatal: not a git repository (or any of the parent directories): .git date09062026@penguin:~$ cd job-automation-engine/
date09062026@penguin:~/job-automation-engine$ git add . date09062026@penguin:~/job-automation-engine$ git commit -m
"date09062026@penguin" On branch main Your branch is up to date with 'origin/main'. nothing to commit, working tree clean
date09062026@penguin:~/job-automation-engine$

```

Response: Here is the entire process consolidated into a single copy-and-paste block.

When your Linux container is recreated, open terminal and paste this whole block:

```

# Silence Crostini notice
echo 10 > ~/.local/share/baguette-motd 2>/dev/null || true

# Update packages and install core build tools
sudo apt update && sudo apt upgrade -y
sudo apt install -y build-essential git curl wget unzip ca-certificates gnupg

# Download and install VS Code
wget 'https://code.visualstudio.com/sha/download?build=stable&os=linux-deb-x64' -O vscode.deb
sudo apt install -y ./vscode.deb
rm vscode.deb

# Configure global Git settings
git config --global user.name "Daniel Blaise Rogers"
git config --global user.email "danielblaiserothers@gmail.com"
git config --global init.defaultBranch main
git config --global core.autocrlf input

# Set up SSH Key for GitHub if not present
if [ ! -f ~/.ssh/id_ed25519 ]; then
    mkdir -p ~/.ssh
    chmod 700 ~/.ssh
    ssh-keygen -t ed25519 -C "danielblaiserothers@gmail.com" -f ~/.ssh/id_ed25519 -N ""
    chmod 600 ~/.ssh/id_ed25519
fi

# Configure ~/.bashrc for SSH Agent
if ! grep -q "Auto-start SSH Agent" ~/.bashrc; then
    cat << 'EOF' >> ~/.bashrc

# Auto-start SSH Agent
if [ -z "$SSH_AUTH_SOCK" ]; then
    eval "$(ssh-agent -s)" > /dev/null
fi

if ! ssh-add -l > /dev/null 2>&1; then
    ssh-add ~/.ssh/id_ed25519 2>/dev/null
fi
EOF
fi

# Reload shell configuration
source ~/.bashrc

# Print public key to add to GitHub
echo "=====
echo "COPY AND ADD THIS SSH PUBLIC KEY TO YOUR GITHUB ACCOUNT:"
echo "=====
cat ~/.ssh/id_ed25519.pub
echo "=====

```


What to do after pasting:

1. Copy the `ssh-ed25519 ...` public key output at the end and add it to **GitHub Settings > SSH and GPG keys**.
2. Clone your repository:

```
git clone git@github.com:YOUR_USERNAME/job-automation-engine.git
cd job-automation-engine
```

3. Re-create your virtual environment and run the backend:

```
python3 -m venv .venv
source .venv/bin/activate
pip install fastapi "uvicorn[standard]" sqlalchemy httpx pydantic-settings python-dotenv
uvicorn app.main:app --reload
```

User prompt: remote: Invalid username or token. Password authentication is not supported for Git operations. fatal: Authentication failed for 'https://github.com/danielblaiserothers/job-automation-engine/'

Response: This error happens because GitHub no longer allows your account password to be used when pushing or pulling over `https://` URLs.

Because you already generated an SSH key (`id_ed25519`) and added it to your GitHub account, the fix is to switch your repository's remote URL from **HTTPS** to **SSH**.

Run these two commands inside your `job-automation-engine` directory:

```
git remote set-url origin git@github.com:danielblaiserothers/job-automation-engine.git
```

Verification: Run the following command to check that your origin URL now starts with `git@github.com`:

```
git remote -v
```

The output should show:

```
origin  git@github.com:danielblaiserothers/job-automation-engine.git (fetch)
origin  git@github.com:danielblaiserothers/job-automation-engine.git (push)
```

Now try pushing or pulling again:

```
git push -u origin main
```

User prompt: how to add a frontend control to the job search

Response: To add a frontend control interface to your job search engine, you can build a lightweight, responsive single-page web interface using HTML, CSS (Tailwind CSS via CDN), and vanilla JavaScript. This frontend allows you to search for jobs by query and location, view the aggregated listings stored in SQLite, and inspect listing details without navigating away from the dashboard.

Step 1: Update CORS in FastAPI (`app/main.py`)

To allow your frontend browser script to communicate with your FastAPI endpoints smoothly, add Cross-Origin Resource Sharing (CORS) middleware in `app/main.py`:

```
from contextlib import asynccontextmanager
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from app.core.config import settings
from app.core.database import init_db
from app.api.v1.endpoints.jobs import router as jobs_router

@asynccontextmanager
async def lifespan(app: FastAPI):
    init_db()
    yield

app = FastAPI(title=settings.PROJECT_NAME, lifespan=lifespan)

# Enable CORS for local development
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
```

```
)
app.include_router(jobs_router, prefix="/api/v1")
```

Step 2: Create the Dashboard File (static/index.html)

Create a new directory named `static` in your project root and add an `index.html` file inside it:

```
mkdir static
touch static/index.html
```

Paste the following HTML/JS control panel code into `static/index.html`:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Job Automation Engine - Control Panel</title>
  <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-gray-900 text-gray-100 min-h-screen font-sans">
  <div class="max-w-6xl mx-auto p-6">

    <!-- Header -->
    <header class="mb-8 border-b border-gray-800 pb-4 flex justify-between items-center">
      <div>
        <h1 class="text-3xl font-bold text-indigo-400">Job Engine Dashboard</h1>
        <p class="text-gray-400 text-sm mt-1">Aggregator, Search Control & Pipeline Tracker</p>
      </div>
      <button onclick="loadStoredJobs()" class="bg-gray-800 hover:bg-gray-700 text-gray-300 px-4 py-2 rounded">
        🔄 Refresh DB
      </button>
    </header>

    <!-- Search Controls -->
    <div class="bg-gray-800 p-6 rounded-lg shadow-lg mb-8 border border-gray-700">
      <h2 class="text-xl font-semibold text-gray-200 mb-4">Fetch New Listings</h2>
      <form id="fetchForm" class="grid grid-cols-1 md:grid-cols-3 gap-4">
        <div>
          <label class="block text-xs font-medium text-gray-400 mb-1">Search Keywords</label>
          <input type="text" id="query" value="python developer" required
            class="w-full bg-gray-900 border border-gray-700 rounded p-2 text-white focus:outline-none fo
          </div>
          <div>
          <label class="block text-xs font-medium text-gray-400 mb-1">Location / Zip Code</label>
          <input type="text" id="location" value="remote" required
            class="w-full bg-gray-900 border border-gray-700 rounded p-2 text-white focus:outline-none fo
          </div>
          <div class="flex items-end">
            <button type="submit" id="submitBtn"
              class="w-full bg-indigo-600 hover:bg-indigo-500 text-white font-medium py-2 px-4 rounded tra
              Fetch & Store Jobs
            </button>
          </div>
        </form>
        <div id="statusMessage" class="mt-3 text-sm font-mono hidden"></div>
      </div>

    <!-- Jobs Table -->
    <div class="bg-gray-800 rounded-lg shadow-lg border border-gray-700 overflow-hidden">
      <div class="p-4 border-b border-gray-700 flex justify-between items-center">
        <h2 class="text-xl font-semibold text-gray-200">Stored Job Listings</h2>
        <span id="jobCount" class="text-xs bg-indigo-900 text-indigo-300 px-2.5 py-1 rounded-full">0 Records</
      </div>
      <div class="overflow-x-auto">
        <table class="w-full text-left text-sm text-gray-300">
          <thead class="bg-gray-900 text-gray-400 uppercase text-xs">
            <tr>
              <th class="py-3 px-4">Title</th>
              <th class="py-3 px-4">Company</th>
              <th class="py-3 px-4">Location</th>
              <th class="py-3 px-4">Source</th>
              <th class="py-3 px-4 text-right">Action</th>
            </tr>
          </thead>
          <tbody id="jobsTableBody" class="divide-y divide-gray-700">
            <tr>
```

```

        <td colspan="5" class="text-center py-6 text-gray-500">No jobs loaded yet. Click "Refresh DB" or
    </tr>
</tbody>
</table>
</div>
</div>
</div>

<script>
    const API_BASE = "http://127.0.0.1:8000/api/v1/jobs";

    // Fetch live jobs from FastAPI DB
    async function loadStoredJobs() {
        try {
            const res = await fetch(`${API_BASE}/`);
            const jobs = await res.json();
            const tbody = document.getElementById("jobsTableBody");
            document.getElementById("jobCount").innerText = `${jobs.length} Records`;

            if (jobs.length === 0) {
                tbody.innerHTML = `<tr><td colspan="5" class="text-center py-6 text-gray-500">No jobs found in datab
                return;
            }

            tbody.innerHTML = jobs.map(job => `
                <tr class="hover:bg-gray-750 transition border-b border-gray-800">
                    <td class="py-3 px-4 font-medium text-white">${job.title}</td>
                    <td class="py-3 px-4 text-gray-300">${job.company}</td>
                    <td class="py-3 px-4 text-gray-400">${job.location}</td>
                    <td class="py-3 px-4"><span class="text-xs bg-gray-700 text-gray-300 px-2 py-0.5 rounded">${job.so
                    <td class="py-3 px-4 text-right">
                        <a href="${job.url}" target="_blank" class="text-indigo-400 hover:text-indigo-300 text-xs font-s
                    </td>
                </tr>
            `).join('');
        } catch (err) {
            console.error("Failed to fetch jobs from database:", err);
        }
    }

    // Trigger API Fetch from Adzuna
    document.getElementById("fetchForm").addEventListener("submit", async (e) => {
        e.preventDefault();
        const query = document.getElementById("query").value;
        const location = document.getElementById("location").value;
        const statusDiv = document.getElementById("statusMessage");
        const submitBtn = document.getElementById("submitBtn");

        statusDiv.classList.remove("hidden");
        statusDiv.className = "mt-3 text-sm font-mono text-yellow-400";
        statusDiv.innerText = "Querying upstream API and processing listings...";
        submitBtn.disabled = true;

        try {
            const res = await fetch(`${API_BASE}/fetch?query=${encodeURIComponent(query)}&location=${encodeURIComponent
                method: 'POST'
            });
            const data = await res.json();

            if (res.ok) {
                statusDiv.className = "mt-3 text-sm font-mono text-green-400";
                statusDiv.innerText = `Success! Added ${data.new_jobs_added} new unique job(s) to SQLite database.`;
                loadStoredJobs();
            } else {
                statusDiv.className = "mt-3 text-sm font-mono text-red-400";
                statusDiv.innerText = `Error: ${data.detail || "Failed to fetch jobs"}`;
            }
        } catch (err) {
            statusDiv.className = "mt-3 text-sm font-mono text-red-400";
            statusDiv.innerText = `Connection Error: ${err.message}`;
        } finally {
            submitBtn.disabled = false;
        }
    });

    // Load stored listings on page boot
    loadStoredJobs();

```

```

</script>
</body>
</html>

```

Step 3: Serve the Static Frontend via FastAPI

To serve this control panel directly from your FastAPI server, mount the static directory in `app/main.py`:

```

from contextlib import asynccontextmanager
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from fastapi.staticfiles import StaticFiles
from fastapi.responses import FileResponse
from app.core.config import settings
from app.core.database import init_db
from app.api.v1.endpoints.jobs import router as jobs_router

@asynccontextmanager
async def lifespan(app: FastAPI):
    init_db()
    yield

app = FastAPI(title=settings.PROJECT_NAME, lifespan=lifespan)

app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

# Register API routes
app.include_router(jobs_router, prefix="/api/v1")

# Serve static dashboard files
app.mount("/static", StaticFiles(directory="static"), name="static")

@app.get("/")
def read_root():
    return FileResponse("static/index.html")

```

Step 4: Run and Test

1. Restart your FastAPI application:


```
uvicorn app.main:app --reload
```
2. Open [<http://127.0.0.1:8000/>] (<http://127.0.0.1:8000/>) in your browser.
3. Use the top control form to submit search queries (e.g., `python developer, remote`), and watch the table update with saved database listings.

User prompt: How to add this method

Response: The **"Error: Method Not Allowed"** occurs because your dashboard script is trying to make an HTTP POST request to `/api/v1/jobs/fetch`, but your backend endpoint in `app/api/v1/endpoints/jobs.py` is currently defined as a `@router.get` (or vice versa).

Align both your backend and frontend to use the same HTTP method.

Method 1: Change the Backend to Accept POST (Recommended)

1. Open `app/api/v1/endpoints/jobs.py`.
2. Make sure the route decorator uses `@router.post`:

```

# app/api/v1/endpoints/jobs.py

@router.post("/fetch")
async def fetch_and_store_jobs(
    query: str = Query(default="python developer"),
    location: str = Query(default="remote"),
    session: Session = Depends(get_session)

```

```

):
  # ... existing fetch code ...

```

Method 2: Change the Dashboard Frontend to Use GET

If you kept `@router.get("/fetch")` in your backend code, update the JavaScript fetch call inside `static/index.html`:

1. Open `static/index.html`.
2. Find the form submission event listener near the bottom of the file.
3. Change method: 'POST' to method: 'GET':

```

// static/index.html

try {
  const res = await fetch(`${API_BASE}/fetch?query=${encodeURIComponent(query)}&location=${encodeURIComponent(
    method: 'GET' // Changed from 'POST' to 'GET'
  )});
  const data = await res.json();
  // ... rest of event listener code

```

Step 3: Test the Fix

1. Restart your FastAPI server:


```
uvicorn app.main:app --reload
```
2. Refresh `http://127.0.0.1:8000/` in your browser.
3. Click **Fetch & Store Jobs**. It will return:


```
Success! Added X new unique job(s) to SQLite database.
```

User prompt: What is a short concise description of what I have built so far?

Response: You have built a modular, local **AI-ready Job Search Aggregator and Application Engine** hosted on a ChromeOS Linux (penguin) container.

Here is a concise breakdown of what is currently operational:

- **Backend Architecture:** A production-structured Python FastAPI application using SQLAlchemy (SQLite) and HTTPX, with clean separation across routing, models, settings, database, and service layers.
- **Live Ingestion Engine:** Asynchronous API client integration with Adzuna that queries active job listings, parses raw JSON payloads, and automatically deduplicates records before storing them in local database tables.
- **Interactive Control Dashboard:** A Tailwind CSS single-page web interface served directly by FastAPI at `localhost:8000`, allowing you to run keyword/location searches, trigger ingestion tasks, refresh records, and view/link to saved jobs.
- **Environment & Version Control:** A fully configured local development stack—including VS Code, Git version control pointing via SSH to GitHub (`job-automation-engine`), automated `ssh-agent` startup routines, and system recovery scripts.

User prompt: <https://support.google.com/voice/answer/10063441?p=not-ringing&rd=1#troubleshoot&zipy=%2Cchrome>

Response: The link you shared points to the Google Support troubleshooter for issues where **Google Voice isn't ringing for incoming calls** when used in Google Chrome.

Here is the step-by-step troubleshooting guide based on Google Voice's official instructions to fix this issue in Chrome:

Step 1: Check Chrome Site Permissions for Google Voice

Google Voice needs explicit permission to play sounds and send notifications in Chrome.

1. Open Google Chrome and go to voice.google.com.
2. Click the **Lock icon** (or Tune/Settings icon) on the left side of the address bar next to `https://voice.google.com`.
3. Ensure the following permissions are set to **Allow**:
 - **Notifications**

- Sound
- Microphone

Step 2: Ensure Google Voice Audio Settings are Configured

Verify that Google Voice is sending ringtones to the correct speaker/headphones.

1. On voice.google.com, click the **Settings** icon (gear icon) in the top-right corner.
2. Select **Audio** on the left menu.
3. Under **Ringling**, click the dropdown and select your primary speakers or headset.
4. Click **Test** next to the selected device to verify you can hear the ringtone clearly.

Step 3: Check Chrome Tab and Browser Mute Settings

- **Unmute Tab:** Right-click the Google Voice browser tab at the top of Chrome and ensure it does not say **Unmute site**. If it does, click it to unmute.
- **Background Activity:** Ensure the Google Voice tab remains open in your browser. Chrome may sleep or throttle inactive tabs if memory-saver settings are enabled.

Step 4: Disable Do Not Disturb

1. Open voice.google.com.
2. Click **Settings** (gear icon) > **Calls**.
3. Under **Do Not Disturb**, ensure **Turn on Do Not Disturb** is toggled **OFF**.

Step 5: Clear Site Cache and Restart Chrome

If the issue persists, cached browser data may be blocking active socket connections:

1. In Chrome, go to `chrome://settings/siteData`.
2. Search for `voice.google.com` and click **Remove All**.
3. Close and reopen Chrome, navigate back to voice.google.com, and log back into your Google account.

User prompt: What did I do?

Response: You resolved an SSH authentication error with GitHub by updating your repository's remote URL from HTTPS to SSH, set up a Tailwind CSS web control panel for your FastAPI job engine, mounted static file serving, and fixed an HTTP 405 "Method Not Allowed" error by aligning your frontend and backend HTTP request methods.

User prompt: NOTICE: To simplify system architecture and maintenance, Crostini has switched by default to a containerless design for new environments starting in ChromeOS version 143 and newer. If you experience unexpected issues with the new design, please report them using the instructions available at <https://www.chromium.org/chromium-os/developer-library/guides/bugs/platform-public-tracker/>. If you would like to revert to the previous system architecture, you may visit `chrome://flags#containerless-crostini` in your Chrome browser and set the flag to "Disabled", then restart your device, and un- and reinstall Crostini. PSA: In order to mitigate the Januscape kernel exploit (see: <https://github.com/V4bel/Januscape>), nested virtualization has been disabled for host kernel versions 5.10 and 5.15. To check your host kernel version, you can navigate to `chrome://system`, find the entry named 'uname', and if it looks something like 'Linux localhost 5.10.XXXX' or 'Linux localhost 5.15.XXX' nested virtualization will be disabled (this message will be repeated 1 more time). (to silence this message, run the following command): `echo 10 >"/home/date09062026/.local/share/baguette-motd"` `date09062026@penguin:~$ cd job-automation-engine/` `date09062026@penguin:~/job-automation-engine$ aconnect venv aconnect - ALSA sequencer connection manager Copyright (C) 1999-2000 Takashi Iwai Usage: * Connection/disconnection between two ports aconnect [-options] sender receiver sender, receiver = client:port pair -d,--disconnect disconnect -e,--exclusive exclusive connection -r,--real # convert real-time-stamp on queue -t,--tick # convert tick-time-stamp on queue * List connected ports (no subscription action) aconnect -i[-o [-options] -i,--input list input (readable) ports -o,--output list output (writable) ports -a,--all show inactive ports, too -l,--list list current connections of each port * Remove all exported connections -x, --removeall date09062026@penguin:~/job-automation-engine$ aconnect venv -x date09062026@penguin:~/job-automation-engine$`

Response: You mistyped the command to activate your Python virtual environment. You ran `aconnect venv` (a Linux MIDI music sequencer utility) instead of `source .venv/bin/activate` or `source venv/bin/activate`.

How to Activate Your Virtual Environment

1. Check the exact name of your virtual environment folder inside `job-automation-engine`:

```
ls -a
```

2. Activate the virtual environment:

- If your folder is named `.venv`:
`source .venv/bin/activate`
- If your folder is named `venv`:
`source venv/bin/activate`

Verification: Your terminal prompt will change to show `(.venv)` or `(venv)` at the very beginning of the line.

If the Virtual Environment Folder Does Not Exist

If the folder was deleted or not created yet, re-create it and install your dependencies:

```
# 1. Create the virtual environment
python3 -m venv .venv

# 2. Activate it
source .venv/bin/activate

# 3. Install required project packages
pip install fastapi "uvicorn[standard]" sqlmodel httpx pydantic-settings python-dotenv

# 4. Start your server
uvicorn app.main:app --reload
```

Silence the MOTD Warning (Optional)

To silence the Crostini notice shown at the top of your terminal, run the command suggested in the banner:

```
mkdir -p ~/.local/share
echo 10 > ~/.local/share/baguette-motd
```

User prompt: help

Response: You are receiving the `{"detail": "Method Not Allowed"}` error (HTTP 405) because typing a URL directly into your browser's address bar sends an HTTP **GET** request, whereas your backend route (`/api/v1/jobs/fetch`) is configured in Python as an HTTP **POST** route.

How to Resolve the Error

Option 1: Trigger the Request from Your Frontend Dashboard

Navigate to your main web interface in your browser:

```
http://127.0.0.1:8000/
```

Use the **Fetch & Store Jobs** button to submit the form via POST.

Option 2: Execute via Interactive API Docs

Go to FastAPI's interactive Swagger documentation:

```
http://127.0.0.1:8000/docs
```

1. Click **POST /api/v1/jobs/fetch**.
2. Click **Try it out**.
3. Fill in your query and location values.
4. Click **Execute**.

Option 3: Change the Backend Decorator to `@router.get`

If you want to trigger job fetches directly by pasting URLs into your browser address bar, update `app/api/v1/endpoints/jobs.py`:

```
# Change this decorator:
# @router.post("/fetch")
```

```
# To this:
@router.get("/fetch")
async def fetch_and_store_jobs(
    query: str = Query(default="python developer"),
    location: str = Query(default="remote"),
    session: Session = Depends(get_session)
):
    ...
```

Save the file and restart your server (`uvicorn app.main:app --reload`). You will then be able to call the URL directly in the browser address bar.

Response: The issue in your `jobs.py` file is that you have defined two functions with the exact same name (`fetch_and_store_jobs`), and the first `@router.get("/fetch")` function is incomplete (it has no function body). In Python, having two functions with the same name causes the second one to overwrite the first.

Here is the clean, corrected version of `app/api/v1/endpoints/jobs.py`. It routes **both** GET and POST requests to a single, complete handler function so you can use browser address bar links (GET) or your dashboard interface (POST) without errors:

```
from fastapi import APIRouter, Depends, HTTPException, Query
from sqlmodel import Session, select
from app.core.database import get_session
from app.models.job import JobListing
from app.services.adzuna_client import AdzunaClient

router = APIRouter(prefix="/jobs", tags=["Jobs"])

# Handle GET and POST on the same endpoint
@router.get("/fetch")
@router.post("/fetch")
async def fetch_and_store_jobs(
    query: str = Query(default=""),
    location: str = Query(default="28716"),
    session: Session = Depends(get_session)
):
    try:
        raw_jobs = await AdzunaClient.fetch_jobs(query=query, location=location)
    except Exception as e:
        raise HTTPException(status_code=500, detail=f"Failed to query upstream API: {str(e)}")

    added_count = 0
    for item in raw_jobs:
        ext_id = str(item.get("id"))

        # Deduplication check
        existing = session.exec(select(JobListing).where(JobListing.external_id == ext_id)).first()
        if not existing:
            job = JobListing(
                external_id=ext_id,
                title=item.get("title", "N/A"),
                company=item.get("company", {}).get("display_name", "Unknown"),
                location=item.get("location", {}).get("display_name", "Remote"),
                description=item.get("description", ""),
                url=item.get("redirect_url", "")
            )
            session.add(job)
            added_count += 1

    session.commit()
    return {"status": "success", "new_jobs_added": added_count}

@router.get("/", response_model=list[JobListing])
def get_jobs(session: Session = Depends(get_session)):
    return session.exec(select(JobListing)).all()
```

Next Steps

1. Replace the contents of `app/api/v1/endpoints/jobs.py` with the code block above.
2. Save the file and restart Uvicorn (`uvicorn app.main:app --reload`).

Response: The file still contains a syntax error because `fetch_and_store_jobs_url` ends abruptly without a body (a `pass` statement, `docstring`, or implementation logic).

Here is the clean, fixed version of `app/api/v1/endpoints/jobs.py`. It uses dual decorators on a single, complete function so that both browser address bar calls (GET) and dashboard POST requests work cleanly without duplicate code or syntax errors:

```
from fastapi import APIRouter, Depends, HTTPException, Query
from sqlmodel import Session, select
from app.core.database import get_session
from app.models.job import JobListing
from app.services.adzuna_client import AdzunaClient

router = APIRouter(prefix="/jobs", tags=["Jobs"])

@router.get("/fetch")
@router.post("/fetch")
async def fetch_and_store_jobs(
    query: str = Query(default=""),
    location: str = Query(default="28716"),
    session: Session = Depends(get_session)
):
    try:
        raw_jobs = await AdzunaClient.fetch_jobs(query=query, location=location)
    except Exception as e:
        raise HTTPException(status_code=500, detail=f"Failed to query upstream API: {str(e)}")

    added_count = 0
    for item in raw_jobs:
        ext_id = str(item.get("id"))

        # Deduplication check
        existing = session.exec(select(JobListing).where(JobListing.external_id == ext_id)).first()
        if not existing:
            job = JobListing(
                external_id=ext_id,
                title=item.get("title", "N/A"),
                company=item.get("company", {}).get("display_name", "Unknown"),
                location=item.get("location", {}).get("display_name", "Remote"),
                description=item.get("description", ""),
                url=item.get("redirect_url", "")
            )
            session.add(job)
            added_count += 1

    session.commit()
    return {"status": "success", "new_jobs_added": added_count}

@router.get("/", response_model=list[JobListing])
def get_jobs(session: Session = Depends(get_session)):
    return session.exec(select(JobListing)).all()
```

Steps to Apply

1. Replace all code in `app/api/v1/endpoints/jobs.py` with the code snippet above.
2. Save the file.
3. Restart Uvicorn: `uvicorn app.main:app --reload`.